INNOPOLIS UNIVERSITY

SNE MASTER TRACK

2025

COURSE: [F25] COMPUTER SYSTEMS AND NETWORKS

CSN LAB 6 REPORT

On

Static Code Analysis

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Static Code Analysis

In this lab your task is simple, you have a list of vulnerable projects and you are required to an automated static code analysis (SAST). Below you will a list of vulnerable applications/projects as well as a list of suggested SAST solutions. pick at least 2 applications/projects and scan them with one of the suggested SAST solution.

Name	URL	Language
DVWA	https://github.com/digininja/	PHP
DVNA	https://github.com/appsecco/dvna	NodeJs
DVJA	https://github.com/appsecco/dvja	Java
InsecureShop	https://github.com/hax0rgb/InsecureShop	Kotlin
dvcsharp-api	https://github.com/appsecco/dvcsharp-api	C#

Name	URL
GitLab SAST	https://docs.gitlab.com/ee/user/application_security/sast/ SonarQube
SonarQube	https://www.sonarqube.org/ https://sonarcloud.io/login
DeepSource	https://deepsource.io/
Snyk	https://snyk.io/
Codiga	https://www.codiga.io/

Solution:

The two projects I chose:

• DVNA (Node.js) - I scanned with Snyk CLI

• DVJA (Java) - I scanned with SonarCloud (sonar-scanner)

I installed these:

Node.js (for Snyk):

First, I opened a browser and went to https://nodejs.org/. From there, I downloaded the LTS version as a Windows Installer (.msi) file. Then, I double-clicked on the downloaded file and followed the Next → Install → Finish steps. After the installation was complete, I opened CMD or Git Bash to check if Node.js was installed correctly.



Java JDK (for SonarScanner + Java project):

I downloaded JDK 11 or above from AdoptOpenJDK / Temurin or Oracle JDK from https://adoptium.net/ website. Then I ran the Windows installer and completed the Next → Install process step by step. After the installation was complete, I opened CMD to confirm whether the Java JDK was installed correctly.

```
C:\Users\Hasib>java -version
openjdk version "25" 2025-09-16 LTS
OpenJDK Runtime Environment Temurin-25+36 (build 25+36-LTS)
OpenJDK 64-Bit Server VM Temurin-25+36 (build 25+36-LTS, mixed mode, sharing)

C:\Users\Hasib>javac -version
javac 25
```

Maven (for Java projects, if the project uses Maven):

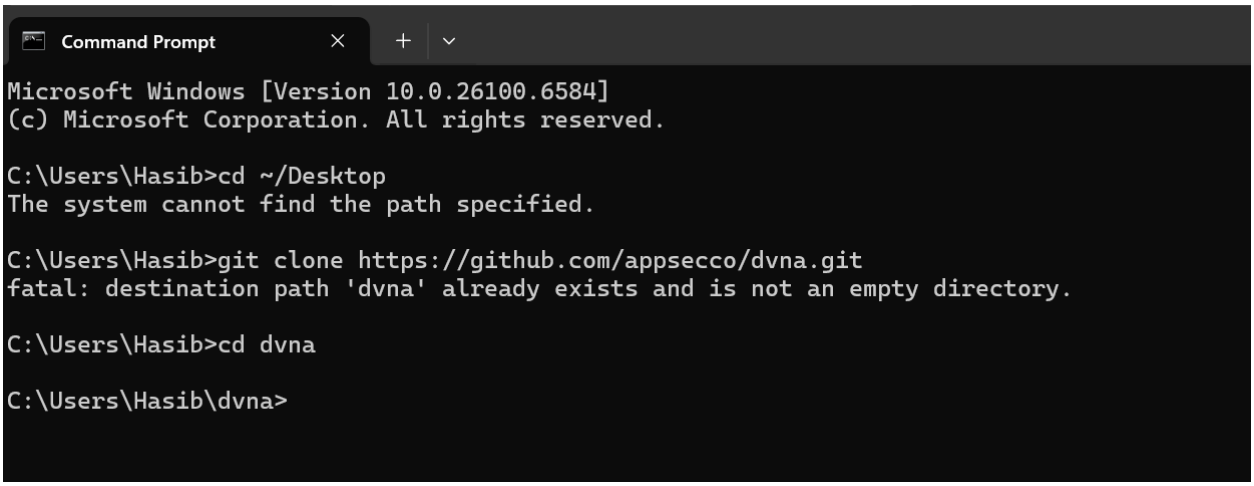
First, I downloaded Maven from the website https://maven.apache.org/download.cgi. I unzipped the downloaded zip file and placed it in the C:\apache-maven\bin folder and added that path to the System Environment Variables (PATH). Then I opened CMD and checked if Maven was installed correctly.

```
C:\Users\Hasib>mvn -v
Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)
Maven home: C:\apache-maven
Java version: 25, vendor: Eclipse Adoptium, runtime: C:\Program Files\Eclipse Adoptium\jdk-25.0.0.36-hotspot
Default locale: en_US, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

DVNA (Node.js) - Scan with Snyk CLI

Project URL: https://github.com/appsecco/dvna

Cloning the project:



Snyk Setup and Login

To install Snyk CLI, I needed Node/npm:

Snyk CLI Install (global):

```
C:\Users\Hasib\dvna>npm install -g snyk
npm warn deprecated boolean@3.2.0: Package no longer supported. Contact Support at https://www.npmjs.com/support for mor
e info.

changed 31 packages in 2m

8 packages are looking for funding
  run `npm fund` for details

C:\Users\Hasib\dvna>
```

Login to Snyk:

The account was seen authenticated.

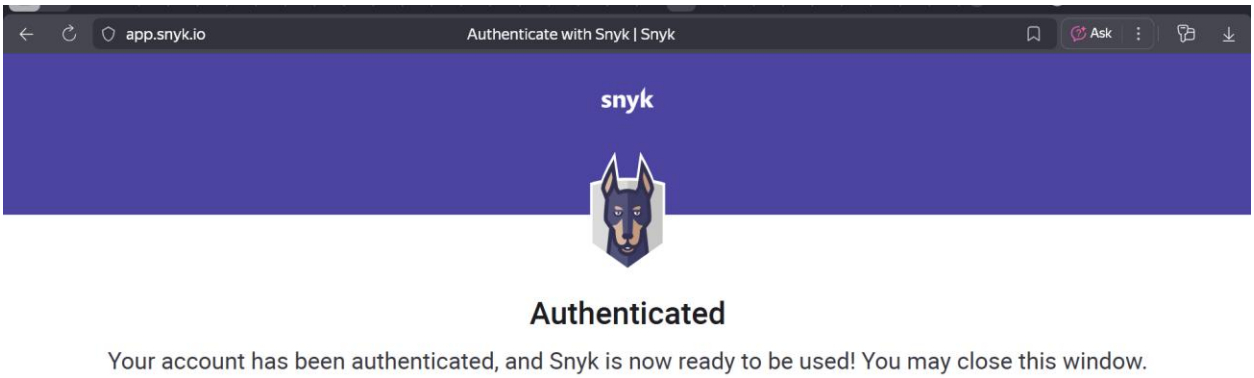
```
Hasib@Hasib MINGW64 ~/Desktop/dvna (master)
$ snyk auth

Now redirecting you to our auth page, go ahead and log in,
and once the auth is complete, return to this prompt and you'll
be ready to start using snyk.

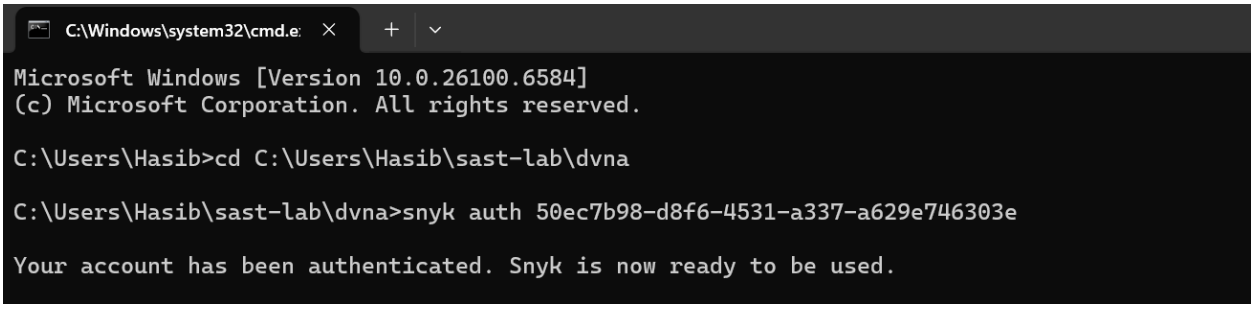
If you can't wait use this url:
https://app.snyk.io/oauth2/authorize?access_type=offline&client_id=b56d4c2e-b9e1
-4d27-8773-ad47eafb0956&code_challenge=BXH6Pmt7CSpKM0t8PQNkb1oap32KaM2EKxTyZbA3L
wM&code_challenge_method=S256&cross_region_routing=true&redirect_uri=http%3A%2F%
2F127.0.0.1%3A8080%2Fauthorization-code%2Fcallback&response_type=code&scope=offl
ine_access&state=Oi1zyWG~I_zUg2S&version=2021-08-11~experimental

Your account has been authenticated.
```

After giving Authentication



The token was used for enabling authentication.

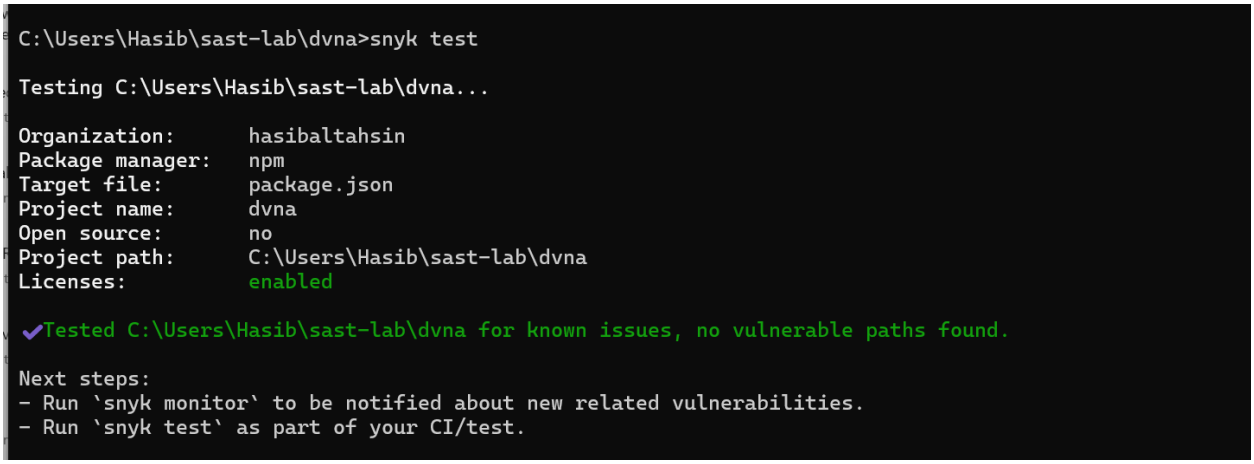


- Typing snyk auth opens your browser and prompts you to sign in/sign up to Snyk.
- To create my Snyk account, I signed up with Sign up with GitHub.
- After signing in, the CLI displays a confirmation message: authenticated as: (user)@domain.com

Running Snyk tests on a project (DVNA — Node.js)

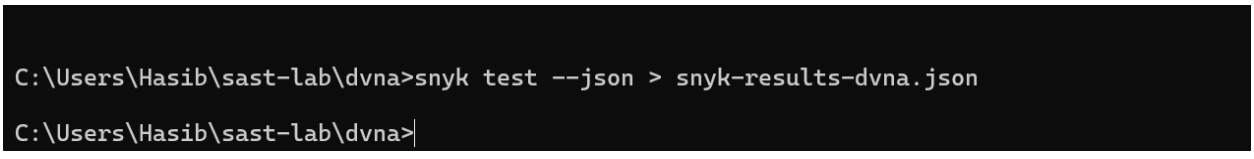
In Git Bash, make sure that the dvna directory (where package.json is located) is:

In the project directory:



The Licenses were showed enabled.

As a snyk report file



This creates the snyk-results-dvna.json file, which can be used to create or submit reports.

DVJA (Java) - Scan with SonarCloud (sonar-scanner)

Project URL: https://github.com/appsecco/dvja

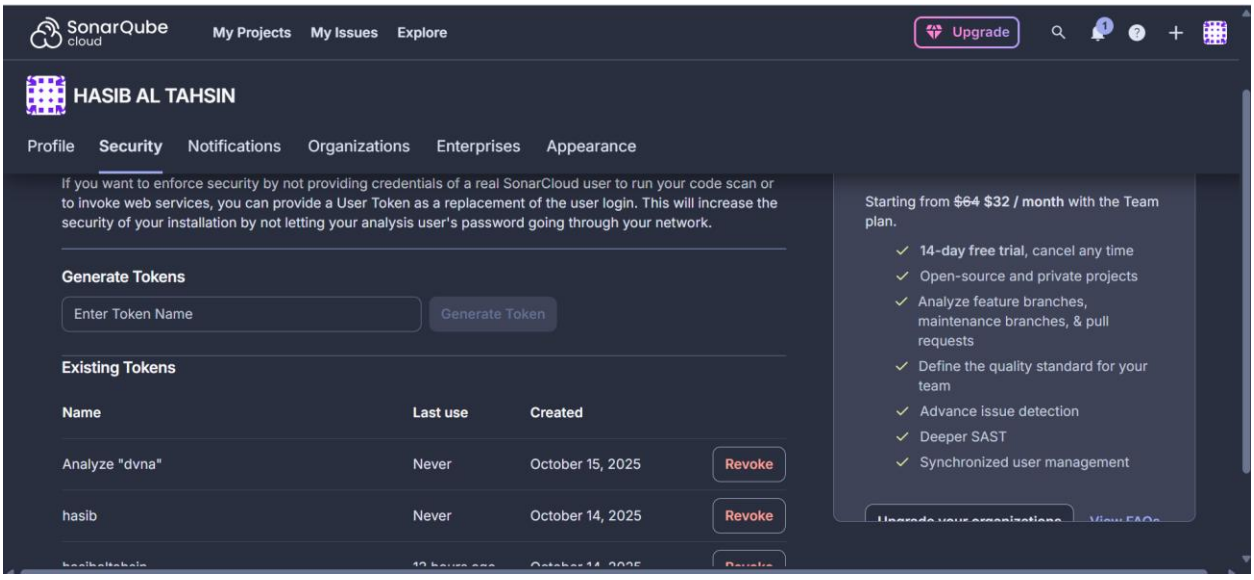
Benefits of using SonarCloud: Online SaaS, SonarCloud reports can be viewed. Results are sent to SonarCloud by running sonar-scanner locally.

Cloning the project:



SonarCloud Account and Token Creation

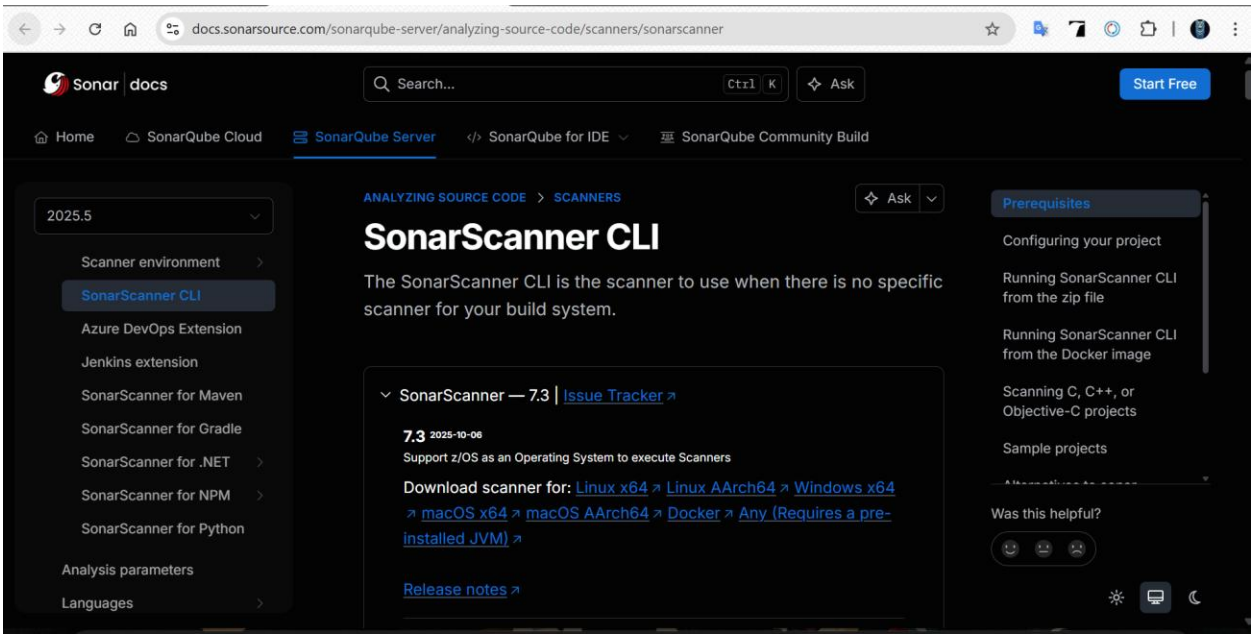
- I Opened browser → https://sonarcloud.io/
- I signed in at the top right → I signed in with GitHub (Authorize with GitHub).
- After logging in, the dashboard appeared. I clicked on the icon at the top right → My Account → Security → Generated Tokens → Gave a name → Generated → I copied the token which could be seen only once. (This token is used later in the CLI)



Download and configure SonarScanner CLI

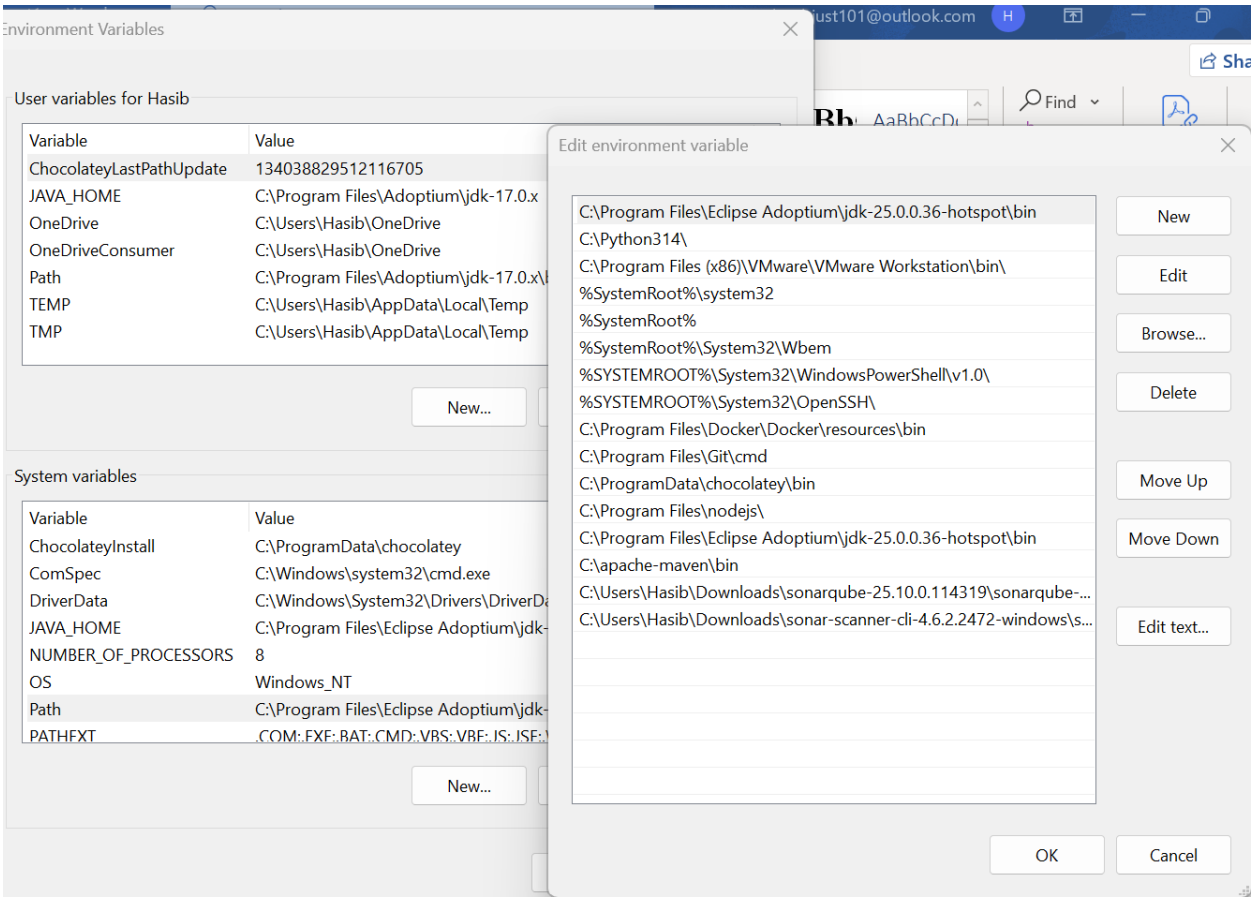
Windows (GUI) installation:

- I opened https://docs.sonarqube.org/latest/analysis/scan/sonarscanner/ (or SonarCloud documentation).
- Then, I downloaded the SonarScanner CLI zip.



Setup on Windows:

- Unzipped the zip and placed it in C:\sonar-scanner\sonar-scanner-<version>.
- Added C:\sonar-scanner\sonar-scanner-<version>\bin to System Properties → Advanced → Environment Variables → Path. Then, I pressed OK.



I ran the Windows .bat file using cmd -c:

I used SonarScanner also:

```
Hasib@Hasib MINGW64 ~/Desktop/dvja (master)
$ cmd -c "C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sona
r-scanner-7.3.0.5189-windows-x64\bin\sonar-scanner.bat"
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Hasib\Desktop\dvja>"C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.51
```

Then I pressed Enter**.**

My sonar-project.properties file was detected (located in the C:\Users\Hasib\Desktop\dvja folder). It then started analyzing the project and finally uploaded the analysis results to SonarCloud under the name hasibaltahsin_dvja.

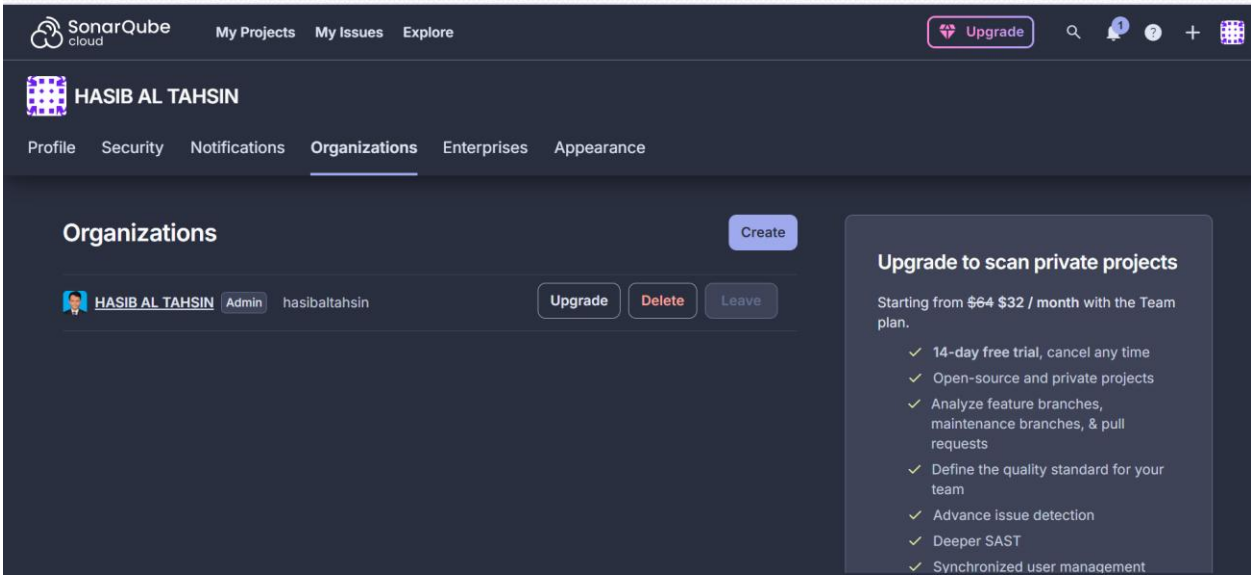
```
C:\Users\Hasib\Desktop\dvja>"C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.51
89-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\bin\sonar-scanner.bat"
15:23:22.540 INFO  Scanner configuration file: C:\Users\Hasib\Downloads\sonar-sc
anner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\bin\..\con
f\sonar-scanner.properties
15:23:22.568 INFO  Project root configuration file: NONE
15:23:22.599 INFO  SonarScanner CLI 7.3.0.5189
15:23:22.604 INFO  Windows 11 10.0 amd64
15:23:24.839 INFO  Communicating with SonarQube Cloud
15:23:24.841 INFO  JRE provisioning: os[windows], arch[amd64]
15:23:25.850 ERROR Failed to query JRE metadata: GET https://api.sonarcloud.io/a
nalysis/jres?os=windows&arch=amd64 failed with HTTP 401. Please check the proper
ty sonar.token or the environment variable SONAR_TOKEN.
15:23:25.851 INFO  EXECUTION FAILURE
15:23:25.852 INFO  Total time: 3.318s

C:\Users\Hasib\Desktop\dvja>
```

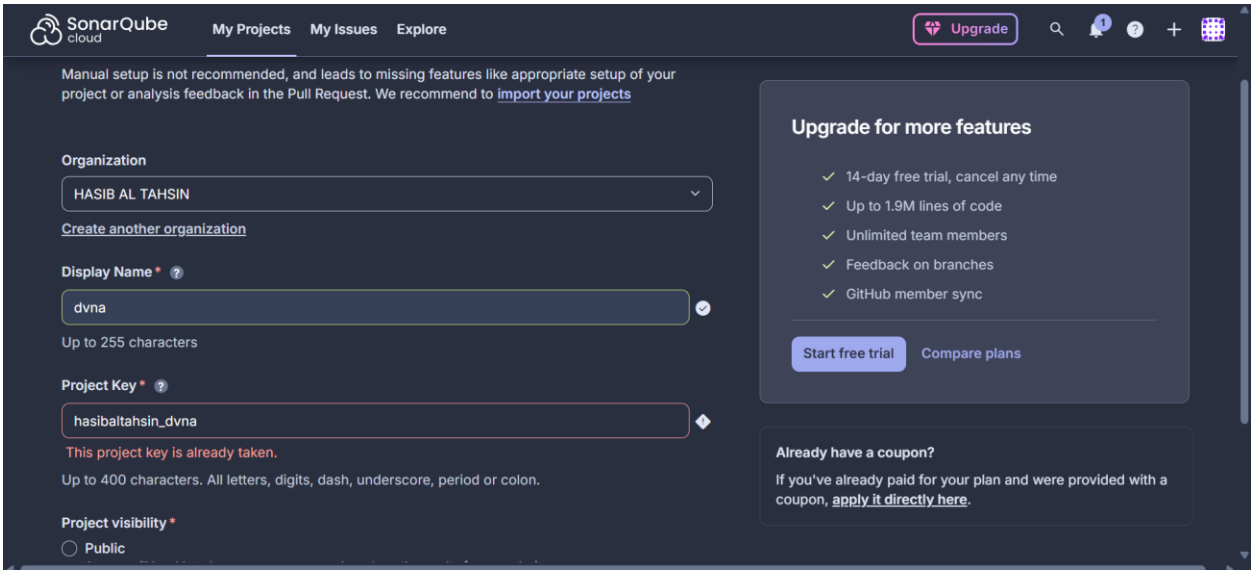
This shows the version.

Creating a project config to create a SonarCloud project:

After entering SonarCloud, first go to the Organizations section. There, select org (if GitHub authentication is done, an organization is automatically created). Then, a new project can be created by clicking on the "+" button or selecting the "Create new project" option.



After selecting the Analyze new project option, I imported the project with a link to my GitHub repository. Giving access to the SonarCloud GitHub App made this process much easier. I saved the project key and name secretly so that I could easily use it in future configurations.



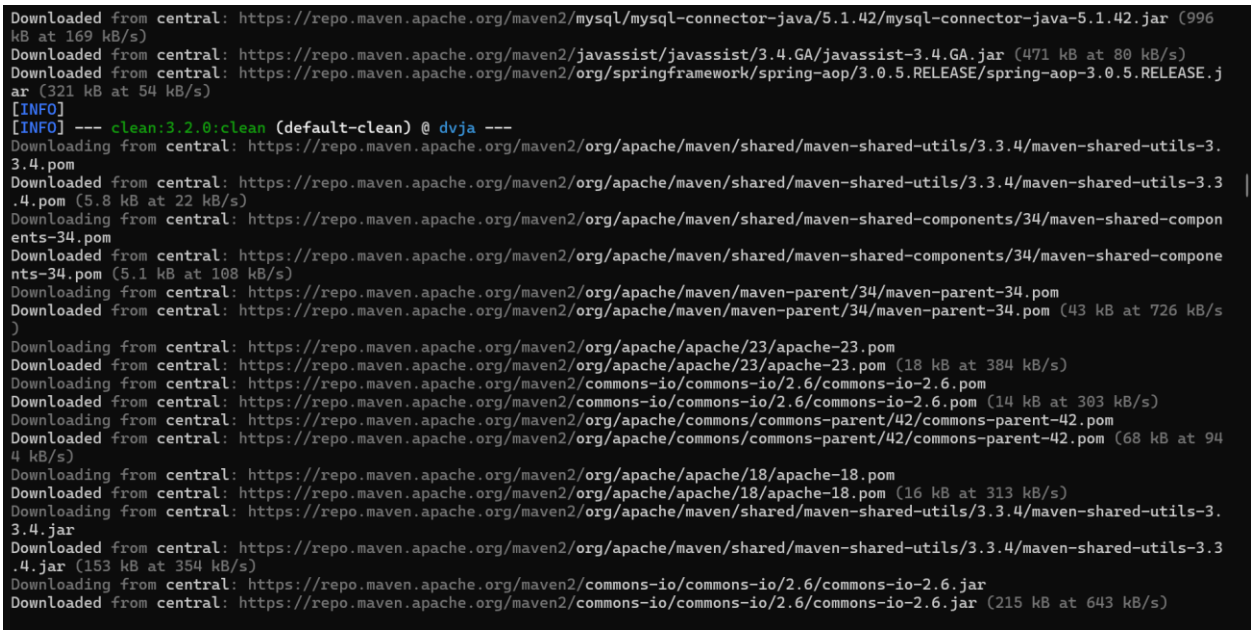
Build your Java project

I used Maven (has pom.xml), so I compiled the project from CMD/PowerShell:

```
C:\Users\Hasib\Desktop\dvja>cd C:\Users\Hasib\Desktop\dvja

C:\Users\Hasib\Desktop\dvja>mvn clean compile
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven
/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.appsecco:dvja >-------------------------
[INFO] Building dvja 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3
.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.
2.0.pom (5.3 kB at 4.6 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom (9.9 kB
at 150 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom (45 kB at 146 kB/s
)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom (21 kB at 285 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3
.2.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.
2.0.jar (36 kB at 440 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.3.1/maven-resources-
plugin-3.3.1.pom
```





This creates .class files in the target/classes folder.

Update sonar-project.properties

I have given the compiled classes path in sonar-project.properties along with the command I gave:

```
File    Edit    View

# sonar-project.properties

sonar.projectKey=hasibaltahsin_dvja
sonar.organization=hasibaltahsin
sonar.host.url=https://sonarcloud.io
sonar.token=16cd549aaeb90330434db9cfaff25a287934c39d
sonar.projectName=DVJA
sonar.language=java
sonar.sources=config,core,docs,models,public,routes,views
```

sonar.java.binaries is my compiled classes path. Maven default is target/classes.

Run SonarScanner

Run from CMD/PowerShell in the project root:

```
C:\Users\Hasib\Desktop\dvja>
C:\Users\Hasib\Desktop\dvja>"C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\bin\sonar-scanner.bat" -X
16:45:37.836 INFO  Scanner configuration file: C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\bin\..\conf\sonar-scanner.properties
16:45:37.847 INFO  Project root configuration file: C:\Users\Hasib\Desktop\dvja\sonar-project.properties
16:45:37.890 INFO  SonarScanner CLI 7.3.0.5189
16:45:37.895 DEBUG Java 17.0.15 Eclipse Adoptium (64-bit)
16:45:37.897 INFO  Windows 11 10.0 amd64
16:45:37.917 DEBUG Scanner max available memory: 3 GB
16:45:37.965 DEBUG Using JVM default truststore: C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\jre\lib\security\cacerts
16:45:37.970 DEBUG Download cache base directory: C:\Users\Hasib\.sonar\cache
16:45:38.206 DEBUG Loading OS trusted SSL certificates...
16:45:38.207 DEBUG This operation might be slow or even get stuck. You can skip it by passing the scanner property 'sonar.scanner.skipSystemTruststore=true'
16:45:38.449 DEBUG Successfully loaded KeyStore of the type [Windows-ROOT] having [58] entries
16:45:38.451 DEBUG Failed to load KeyStore of the type [Windows-ROOT-LOCALMACHINE]
16:45:38.534 DEBUG Successfully loaded KeyStore of the type [Windows-ROOT-CURRENTUSER] having [58] entries
16:45:38.573 DEBUG Successfully loaded KeyStore of the type [Windows-MY] having [4] entries
16:45:38.596 DEBUG Successfully loaded KeyStore of the type [Windows-MY-CURRENTUSER] having [4] entries
16:45:38.598 DEBUG Failed to load KeyStore of the type [Windows-MY-LOCALMACHINE]
16:45:38.602 DEBUG Loaded [124] system trusted certificates
16:45:39.092 DEBUG Loaded truststore from 'C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\jre\lib\security\cacerts' containing 150 certificates
16:45:39.725 INFO  Communicating with SonarQube Cloud
16:45:39.728 INFO  JRE provisioning: os[windows], arch[amd64]
16:45:39.860 DEBUG --> GET https://api.sonarcloud.io/analysis/jres?os=windows&arch=amd64
16:45:41.532 DEBUG <-- 200 https://api.sonarcloud.io/analysis/jres?os=windows&arch=amd64 (1670ms, 475-byte body)
16:45:41.621 DEBUG Executing: C:\Users\Hasib\.sonar\cache\4bafe2e94439c8193fc8c68247cb0dbaf4e80265b903288f63f128304f129bbe\OpenJDK17U-jre_x64_windows_hotspot_17.0.11_9.zip_extracted\jdk-17.0.11+9-jre\bin\java.exe --version
16:45:41.798 DEBUG openjdk 17.0.11 2024-04-16
16:45:41.799 DEBUG OpenJDK Runtime Environment Temurin-17.0.11+9 (build 17.0.11+9)
```

```
-windows-x64\jre\lib\security\cacerts' containing 150 certificates
16:45:47.809 DEBUG User cache: C:\Users\Hasib\.sonar\cache
16:45:47.840 INFO  Loading required plugins
16:45:47.842 INFO  Load plugins index
16:45:47.847 DEBUG --> GET https://sonarcloud.io/api/plugins/installed
16:45:48.230 DEBUG <-- 200 https://sonarcloud.io/api/plugins/installed (382ms, unknown-length body)
16:45:48.248 INFO  Load plugins index (done) | time=407ms
16:45:48.250 INFO  Load/download plugins
16:45:48.379 INFO  Load/download plugins (done) | time=130ms
16:45:48.380 DEBUG Plugins not loaded because they are optional: [abap, sonarapex, architecture, architecturejavascriptfrontend, arch
itecturepythonfrontend, architecturecsharpfrontend, architecturejavafrontend, csharpenterprise, cpp, cobol, dart, dbd, dbdjavafronten
drewrite, dbdpythonfrontend, dre, flex, go, web, jcl, javasymbolicexecution, java, javascript, kotlin, php, pli, plsql, python, rpg,
ruby, rust, sonarscala, jasmin, swift, tsql, vbnetenterprise, vb, security, securitycsharpfrontend, securitygofrontend, securityjsfro
ntend, securityjavafrontend, securitykotlinfrontend, securityphpfrontend, securitypythonfrontend, securityvbnetfrontend, xml]
16:45:48.506 DEBUG Plugins loaded:
16:45:48.507 DEBUG   * Text Code Quality and Security 2.30.0.8328 (textenterprise)
16:45:48.508 DEBUG   * JaCoCo 1.3.0.1538 (jacoco)
16:45:48.509 DEBUG   * IaC Code Quality and Security 1.51.0.16714 (iacenterprise)
16:45:48.511 DEBUG   * IaC Code Quality and Security 1.51.0.16714 (iac)
16:45:48.512 DEBUG --> GET https://sonarcloud.io/api/server/version
16:45:48.880 DEBUG <-- 200 https://sonarcloud.io/api/server/version (370ms, 11-byte body)
16:45:48.884 DEBUG Updated analysis started with a difference of 320 milliseconds
16:45:48.910 DEBUG Started at Wed Oct 15 16:45:48 BDT 2025
16:45:49.461 DEBUG register org.eclipse.jgit.util.FS$FileStoreAttributes$$Lambda$268/0x00000222812eb958@34819867 with shutdown hook
16:45:49.514 DEBUG Could not detect any CI vendor
16:45:49.544 INFO  Load project settings for component key: 'hasibaltahsin_dvja'
16:45:49.545 DEBUG --> GET https://sonarcloud.io/api/settings/values.protobuf?component=hasibaltahsin_dvja
16:45:50.143 DEBUG <-- 200 https://sonarcloud.io/api/settings/values.protobuf?component=hasibaltahsin_dvja (597ms, unknown-length bod
y)
16:45:50.157 INFO  Load project settings for component key: 'hasibaltahsin_dvja' (done) | time=612ms
16:45:50.185 INFO  Process project properties
16:45:50.220 DEBUG Process project properties (done) | time=35ms
16:45:50.221 INFO  Execute project builders
```

```
16:45:56.966 DEBUG Declared patterns of language XML were converted to sonar.lang.patterns.xml : **/*.xml,**/*.xsd,**/*.xsl,**/*.conf
ig
16:45:56.967 DEBUG Declared patterns of language JSON were converted to sonar.lang.patterns.json : **/*.json
16:45:56.968 DEBUG Declared patterns of language IPython Notebooks were converted to sonar.lang.patterns.ipynb : **/*.ipynb
16:45:56.969 DEBUG Declared patterns of language Text were converted to sonar.lang.patterns.text :
16:45:56.969 DEBUG Declared patterns of language VB.NET were converted to sonar.lang.patterns.vbnet : **/*.vb
16:45:56.970 DEBUG Declared patterns of language CloudFormation were converted to sonar.lang.patterns.cloudformation :
16:45:56.970 DEBUG Declared patterns of language Swift were converted to sonar.lang.patterns.swift : **/*.swift
16:45:56.971 DEBUG Declared patterns of language YAML were converted to sonar.lang.patterns.yaml : **/*.yaml,**/*.yml
16:45:56.972 DEBUG Declared patterns of language C++ were converted to sonar.lang.patterns.cpp : **/*.cc,**/*.cpp,**/*.cxx,**/*.c++,*
*/*.hh,**/*.hpp,**/*.hxx,**/*.h++,**/*.ipp,**/*.ixx,**/*.mxx,**/*.cppm,**/*.ccm,**/*.cxxm,**/*.c++m
16:45:56.972 DEBUG Declared patterns of language C were converted to sonar.lang.patterns.c : **/*.c,**/*.h
16:45:56.973 DEBUG Declared patterns of language Go were converted to sonar.lang.patterns.go : **/*.go
16:45:56.974 DEBUG Declared patterns of language Kotlin were converted to sonar.lang.patterns.kotlin : **/*.kt,**/*.kts
16:45:56.975 DEBUG Declared patterns of language RPG were converted to sonar.lang.patterns.rpg : **/*.rpg,**/*.rpgle,**/*.sqlrpgle,**
/*.rpg,**/*.rpgle,**/*.sqlrpgle
16:45:56.975 DEBUG Declared patterns of language PL/I were converted to sonar.lang.patterns.pli : **/*.pli
16:45:56.976 DEBUG Declared patterns of language T-SQL were converted to sonar.lang.patterns.tsql : **/*.tsql
16:45:56.976 DEBUG Declared patterns of language Vb were converted to sonar.lang.patterns.vb : **/*.bas,**/*.frm,**/*.ctl
16:45:56.977 DEBUG Declared patterns of language Secrets were converted to sonar.lang.patterns.secrets :
16:45:56.978 DEBUG Declared patterns of language Ruby were converted to sonar.lang.patterns.ruby : **/*.rb
16:45:56.978 DEBUG Declared patterns of language C# were converted to sonar.lang.patterns.cs : **/*.cs,**/*.razor
16:45:56.979 DEBUG Declared patterns of language COBOL were converted to sonar.lang.patterns.cobol :
16:45:56.982 DEBUG Declared patterns of language Shell were converted to sonar.lang.patterns.shell : **/*.sh,**/*.bash
16:45:56.983 DEBUG Declared patterns of language PHP were converted to sonar.lang.patterns.php : **/*.php,**/*.php3,**/*.php4,**/*.ph
p5,**/*.phtml,**/*.inc
16:45:56.984 DEBUG Declared patterns of language Terraform were converted to sonar.lang.patterns.terraform : **/*.tf
16:45:56.984 DEBUG Declared patterns of language Azure Resource Manager were converted to sonar.lang.patterns.azureresourcemanager :
**/*.bicep
16:45:56.985 DEBUG Declared patterns of language ABAP were converted to sonar.lang.patterns.abap : **/*.abap,**/*.ab4,**/*.flow,**/*.
asprog
16:45:56.985 DEBUG Declared patterns of language GitHub Actions were converted to sonar.lang.patterns.githubactions :
16:45:56.986 DEBUG Declared patterns of language Objective-C were converted to sonar.lang.patterns.objc : **/*.m
```

```
16:45:57.358 DEBUG git-lfs filter-processfile:C:/Program Files/Git/etc/gitconfigfilter.lfs.required
16:45:57.359 DEBUG truefile:C:/Program Files/Git/etc/gitconfighttp.sslbackend
16:45:57.359 DEBUG schannelfile:C:/Program Files/Git/etc/gitconfigcore.autocrlf
16:45:57.360 DEBUG truefile:C:/Program Files/Git/etc/gitconfigcore.fscache
16:45:57.361 DEBUG truefile:C:/Program Files/Git/etc/gitconfigcore.symlinks
16:45:57.362 DEBUG falsefile:C:/Program Files/Git/etc/gitconfigpull.rebase
16:45:57.363 DEBUG falsefile:C:/Program Files/Git/etc/gitconfigcredential.helper
16:45:57.364 DEBUG managerfile:C:/Program Files/Git/etc/gitconfigcredential.https://dev.azure.com.usehttppath
16:45:57.364 DEBUG truefile:C:/Program Files/Git/etc/gitconfiginit.defaultbranch
16:45:57.367 DEBUG master
16:45:57.370 DEBUG loading config FileBasedConfig[C:\Program Files\Git\etc\gitconfig]
16:45:57.373 DEBUG loading config FileBasedConfig[C:\Users\Hasib\.config\git\config]
16:45:57.374 DEBUG loading config UserConfigFile[C:\Users\Hasib\.gitconfig]
16:45:57.530 DEBUG 178 non excluded files in this Git repository
16:45:58.125 INFO  7 languages detected in 160 preprocessed files (done) | time=1092ms
16:45:58.126 INFO  0 files ignored because of inclusion/exclusion patterns
16:45:58.127 INFO  0 files ignored because of scm ignore settings
16:45:58.481 INFO  Loading plugins for detected languages
16:45:58.481 DEBUG Detected languages: [css, java, shell, web, jsp, xml, js]
16:45:58.482 INFO  Load/download plugins
16:45:58.822 INFO  Load/download plugins (done) | time=340ms
16:45:58.824 DEBUG Optional language-specific plugins not loaded: [abap, sonarapex, architecturepythonfrontend, architecturecsharpfro
ntend, csharpenterprise, cpp, cobol, dart, dbdpythonfrontend, flex, go, jcl, kotlin, php, pli, plsql, python, rpg, ruby, rust, sonars
cala, swift, tsql, vbnetenterprise, vb, securitycsharpfrontend, securitygofrontend, securitykotlinfrontend, securityphpfrontend, secu
ritypythonfrontend, securityvbnetfrontend]
16:45:59.045 DEBUG Plugins loaded:
16:45:59.045 DEBUG   * Dataflow Bug Detection Rules for Java 2.4.0.18695 (dbdjavafrontendrewrite)
16:45:59.046 DEBUG   * Dataflow Bug Detection 2.4.0.18695 (dbd)
16:45:59.047 DEBUG   * JavaScript/TypeScript/CSS Code Quality and Security 11.4.1.34873 (javascript)
16:45:59.048 DEBUG   * Vulnerability Rules for Java 11.10.0.40378 (securityjavafrontend)
16:45:59.048 DEBUG   * Vulnerability Analysis 11.10.0.40378 (security)
16:45:59.049 DEBUG   * Vulnerability Rules for JS 11.10.0.40378 (securityjsfrontend)
16:45:59.050 DEBUG   * Java Code Quality and Security 8.19.0.40387 (java)
```

```
16:46:01.192 DEBUG 'src/main/webapp/WEB-INF/dvja/a9_vuln_component/reference.jsp' indexed with language 'jsp'
16:46:01.193 DEBUG 'src/main/webapp/WEB-INF/dvja/a9_vuln_component/scenario.jsp' indexed with language 'jsp'
16:46:01.195 DEBUG 'src/main/webapp/WEB-INF/dvja/a9_vuln_component.jsp' indexed with language 'jsp'
16:46:01.196 DEBUG 'src/main/webapp/WEB-INF/dvja/AssessmentHome.jsp' indexed with language 'jsp'
16:46:01.197 DEBUG 'src/main/webapp/WEB-INF/dvja/common/Footer.jsp' indexed with language 'jsp'
16:46:01.197 DEBUG 'src/main/webapp/WEB-INF/dvja/common/Head.jsp' indexed with language 'jsp'
16:46:01.197 DEBUG 'src/main/webapp/WEB-INF/dvja/common/Menu.jsp' indexed with language 'jsp'
16:46:01.198 DEBUG 'src/main/webapp/WEB-INF/dvja/common/Navigation.jsp' indexed with language 'jsp'
16:46:01.199 DEBUG 'src/main/webapp/WEB-INF/dvja/Home.jsp' indexed with language 'jsp'
16:46:01.200 DEBUG 'src/main/webapp/WEB-INF/dvja/Login.jsp' indexed with language 'jsp'
16:46:01.200 DEBUG 'src/main/webapp/WEB-INF/dvja/Ping.jsp' indexed with language 'jsp'
16:46:01.200 DEBUG 'src/main/webapp/WEB-INF/dvja/ProductEdit.jsp' indexed with language 'jsp'
16:46:01.201 DEBUG 'src/main/webapp/WEB-INF/dvja/ProductList.jsp' indexed with language 'jsp'
16:46:01.202 DEBUG 'src/main/webapp/WEB-INF/dvja/Registration.jsp' indexed with language 'jsp'
16:46:01.202 DEBUG 'src/main/webapp/WEB-INF/dvja/RegistrationSuccess.jsp' indexed with language 'jsp'
16:46:01.202 DEBUG 'src/main/webapp/WEB-INF/dvja/ResetPasswordInput.jsp' indexed with language 'jsp'
16:46:01.203 DEBUG 'src/main/webapp/WEB-INF/dvja/ResetPasswordStart.jsp' indexed with language 'jsp'
16:46:01.203 DEBUG 'src/main/webapp/WEB-INF/dvja/ResetPasswordSuccess.jsp' indexed with language 'jsp'
16:46:01.204 DEBUG 'src/main/webapp/WEB-INF/dvja/UserEdit.jsp' indexed with language 'jsp'
16:46:01.204 DEBUG 'src/main/webapp/WEB-INF/dvja/UserSearch.jsp' indexed with language 'jsp'
16:46:01.206 DEBUG 'src/main/webapp/WEB-INF/example/HelloWorld.jsp' indexed with language 'jsp'
16:46:01.209 DEBUG 'src/main/webapp/WEB-INF/example/Login.jsp' indexed with language 'jsp'
16:46:01.210 DEBUG 'src/main/webapp/WEB-INF/example/Menu.jsp' indexed with language 'jsp'
16:46:01.213 DEBUG 'src/main/webapp/WEB-INF/example/Missing.jsp' indexed with language 'jsp'
16:46:01.214 DEBUG 'src/main/webapp/WEB-INF/example/Register.jsp' indexed with language 'jsp'
16:46:01.214 DEBUG 'src/main/webapp/WEB-INF/example/Welcome.jsp' indexed with language 'jsp'
16:46:01.215 DEBUG 'src/main/webapp/WEB-INF/web.xml' indexed with language 'xml'
16:46:01.215 DEBUG 'docs/.gitignore' indexed with language 'null'
16:46:01.216 DEBUG 'docs/assessment/a1-injection.md' indexed with language 'null'
16:46:01.217 DEBUG 'docs/assessment/a3-cross-site-scripting.md' indexed with language 'null'
16:46:01.218 DEBUG 'docs/assessment/a4-insecure-direct-object-reference.md' indexed with language 'null'
16:46:01.219 DEBUG 'docs/assets/cmdi.png' indexed with language 'null'
16:46:01.220 DEBUG 'docs/assets/intellij_hw.png' indexed with language 'null'
```

```
16:46:01.250 INFO  160 files indexed (done) | time=140ms
16:46:01.269 DEBUG Available languages:
16:46:01.270 DEBUG   * Secrets => "secrets"
16:46:01.270 DEBUG   * Text => "text"
16:46:01.271 DEBUG   * Ansible => "ansible"
16:46:01.273 DEBUG   * GitHub Actions => "githubactions"
16:46:01.273 DEBUG   * Terraform => "terraform"
16:46:01.274 DEBUG   * CloudFormation => "cloudformation"
16:46:01.274 DEBUG   * Kubernetes => "kubernetes"
16:46:01.274 DEBUG   * Docker => "docker"
16:46:01.275 DEBUG   * Azure Resource Manager => "azureresourcemanager"
16:46:01.275 DEBUG   * YAML => "yaml"
16:46:01.275 DEBUG   * JSON => "json"
16:46:01.276 DEBUG   * JavaScript => "js"
16:46:01.276 DEBUG   * TypeScript => "ts"
16:46:01.277 DEBUG   * CSS => "css"
16:46:01.278 DEBUG   * Java => "java"
16:46:01.279 DEBUG   * HTML => "web"
16:46:01.280 DEBUG   * JSP => "jsp"
16:46:01.281 DEBUG   * Shell => "shell"
16:46:01.281 DEBUG   * XML => "xml"
16:46:01.282 INFO  Quality profile for css: Sonar way
16:46:01.282 INFO  Quality profile for java: Sonar way
16:46:01.282 INFO  Quality profile for js: Sonar way
16:46:01.283 INFO  Quality profile for jsp: Sonar way
16:46:01.283 INFO  Quality profile for shell: Sonar way
16:46:01.284 INFO  Quality profile for web: Sonar way
16:46:01.284 INFO  Quality profile for xml: Sonar way
16:46:01.285 INFO  ------------- Run sensors on module DVJA
16:46:01.497 INFO  SonarJasmin relies on SonarSecurity to define the JS/TS security rule repositories.
16:46:01.528 DEBUG Scanner for generating DBD IR instantiated.
16:46:02.426 DEBUG Added 392 checks for language='ts', repository='typescript'
16:46:02.448 DEBUG Added 15 checks for language='ts', repository='tssecurity'
```

```
:S6384), ReflectionInjectionCheck (javasecurity:S6173), NoSQLInjectionCheck (javasecurity:S5147), XMLOperationsCheck (javasecurity:S6
399), JSONOperationsCheck (javasecurity:S6398), ThreadSuspensionDoSCheck (javasecurity:S6390), FileSystemOracleCheck (javasecurity:S6
549), OSEnvironmentVariablesCheck (javasecurity:S6547), ApiTraversalCheck (javasecurity:S7044), PromptInjectionCheck (javasecurity:S7
518), CrossApplicationAttackCheck (javasecurity:S7606), SensitiveInformationDisclosureCheck (javasecurity:S7610)]
16:46:02.888 DEBUG Registered check: []
16:46:03.046 DEBUG Registered check: [RedundantAssignmentsCheck (java:S4165), MinMaxRangeCheck (java:S3065), XmlValidatedSignatureChe
ck (java:S6377), ConditionalUnreachableCodeCheck (java:S2583), DenialOfServiceXMLCheck (java:S6376), AllowXMLInclusionCheck (java:S63
73), UnclosedResourcesCheck (java:S2095), NonNullSetToNullCheck (java:S2637), InvariantReturnCheck (java:S3516), MapComputeIfAbsentOr
PresentCheck (java:S3824), StreamConsumedCheck (java:S3959), StreamNotConsumedCheck (java:S3958), OptionalGetBeforeIsPresentCheck (ja
va:S3655), BooleanGratuitousExpressionsCheck (java:S2589), XxeProcessingCheck (java:S2755), ObjectOutputStreamCheck (java:S2689), Par
ameterNullnessCheck (java:S4449), LocksNotUnlockedCheck (java:S2222), NoWayOutLoopCheck (java:S2189)]
16:46:04.394 DEBUG Registered check: [AbsOnNegativeCheck (java:S2676), AccessibilityChangeCheck (java:S3011), AccessibilityChangeOnRe
cordsCheck (java:S6216), AllBranchesAreIdenticalCheck (java:S3923), AnchorPrecedenceCheck (java:S5850), AndroidBiometricAuthWithoutCr
yptoCheck (java:S6293), AndroidBroadcastingCheck (java:S5320), AndroidExternalStorageCheck (java:S5324), AndroidMobileDatabaseEncrypt
ionKeysCheck (java:S6301), AndroidNonAuthenticatedUsersCheck (java:S6288), AndroidPersistentUniqueIdentifierCheck (java:S7435), Anony
mousClassShouldBeLambdaCheck (java:S1604), ArrayCopyLoopCheck (java:S3012), ArrayDesignatorAfterTypeCheck (java:S1195), ArrayDesignat
orOnVariableCheck (java:S1197), ArrayForVarArgCheck (java:S3878), ArrayHashCodeAndToStringCheck (java:S2116), ArraysAsListOfPrimitive
ToStreamCheck (java:S3631), AssertOnBooleanVariableCheck (java:S3346), AssertionsInProductionCodeCheck (java:S5960), AssertsOnParamet
ersOfPublicMethodCheck (java:S4274), AssignmentInSubExpressionCheck (java:S1121), AsyncMethodsCalledViaThisCheck (java:S6809), AsyncM
ethodsOnConfigurationClassCheck (java:S6817), AsyncMethodsReturnTypeCheck (java:S6810), AuthorizationsStrongDecisionsCheck (java:S580
8), AutowiredOnConstructorWhenMultipleConstructorsCheck (java:S6829), AutowiredOnMultipleConstructorsCheck (java:S6818), AvoidQualifi
erOnBeanMethodsCheck (java:S6831), AwsConsumerBuilderUsageCheck (java:S6244), AwsCredentialsShouldBeSetExplicitlyCheck (java:S6242),
AwsLambdaSyncCallCheck (java:S6246), AwsLongTermAccessKeysCheck (java:S6263), AwsRegionSetterCheck (java:S6262), AwsRegionShouldBeSet
ExplicitlyCheck (java:S6241), AwsReusableResourcesInitializedOnceCheck (java:S6243), BadClassNameCheck (java:S101), BadConstantNameCh
eck (java:S115), BadFieldNameCheck (java:S116), BadFieldNameStaticNonFinalCheck (java:S3008), BadInterfaceNameCheck (java:S114), BadL
ocalVariableNameCheck (java:S117), BadMethodNameCheck (java:S100), BadPackageNameCheck (java:S120), BadTypeParameterNameCheck (java:S
119), BatchSQLStatementsCheck (java:S6912), BigDecimalDoubleConstructorCheck (java:S2111), BlockingOperationsInVirtualThreadsCheck (j
ava:S6881), BooleanInversionCheck (java:S1940), BooleanLiteralCheck (java:S1125), BooleanMethodReturnCheck (java:S2447), BoxedBoolean
ExpressionsCheck (java:S5411), BrainMethodCheck (java:S6541), CORSCheck (java:S5122), CacheAnnotationsShouldOnlyBeAppliedToConcreteCl
assesCheck (java:S7180), CallOuterPrivateMethodCheck (java:S3398), CallSuperMethodFromInnerClassCheck (java:S2388), CallToDeprecatedC
odeMarkedForRemovalCheck (java:S5738), CallToDeprecatedMethodCheck (java:S1874), CanonEqFlagInRegexCheck (java:S5854), CaseInsensitiv
eComparisonCheck (java:S1157), CastArithmeticOperandCheck (java:S2184), CatchIllegalMonitorStateExceptionCheck (java:S2235), CatchOfT
hrowableOrErrorCheck (java:S1181), CatchRethrowingCheck (java:S2737), ChangeMethodContractCheck (java:S2638), ChildClassShadowFieldCh
```

```
sCheck (java:S5961), TryWithResourcesCheck (java:S2093), UnnamedVariableShouldUseVarCheck (java:S7466), UnusedCollectionCheck (java:S
4030), UnusedLocalVariableCheck (java:S1481), UnusedMethodParameterCheck (java:S1172), UnusedPrivateFieldCheck (java:S1068), UnusedPr
ivateMethodCheck (java:S1144), UnusedTestRuleCheck (java:S2924), UseOfSequentialForSequentialGathererCheck (java:S7481), UsePageableP
arameterForPagedQueryCheck (java:S7186), UseTransformClassInsteadOfBuildCheck (java:S7478), UselessImportCheck (java:S1128), UselessP
arenthesesCheck (java:S1110), WebViewJavaScriptInterfaceCheck (java:S7409)]
16:46:04.764 DEBUG 'Import external issues report' skipped because of missing configuration requirements.
Accessed configuration:
- sonar.externalIssuesReportPaths: <empty>
16:46:04.767 DEBUG 'IaC Ansible Sensor' skipped because there is no related file in current project
16:46:04.770 DEBUG 'IaC ansible-lint report Sensor' skipped because there is no related file in current project
16:46:04.772 DEBUG 'IaC Terraform Sensor' skipped because there is no related file in current project
16:46:04.773 DEBUG 'IaC TFLint report Sensor' skipped because there is no related file in current project
16:46:04.775 DEBUG 'IaC CloudFormation Sensor' skipped because there is no related file in current project
16:46:04.776 DEBUG 'IaC cfn-lint report Sensor' skipped because there is no related file in current project
16:46:04.779 DEBUG 'IaC Kubernetes Sensor' skipped because there is no related file in current project
16:46:04.782 DEBUG 'IaC Azure Resource Manager Sensor' skipped because there is no related file in current project
16:46:04.783 DEBUG 'JavaScript inside YAML analysis' skipped because there is no related file in current project
16:46:04.785 DEBUG 'JavaScript/TypeScript Coverage' skipped because of missing configuration requirements.
Accessed configuration:
- sonar.javascript.lcov.reportPaths: <empty>
- sonar.externalIssuesReportPaths: <empty>
- sonar.typescript.lcov.reportPaths: <empty>
16:46:04.787 DEBUG 'Import of TSLint issues' skipped because of missing configuration requirements.
Accessed configuration:
- sonar.typescript.tslint.reportPaths: <empty>
- sonar.javascript.lcov.reportPaths: <empty>
- sonar.externalIssuesReportPaths: <empty>
- sonar.typescript.lcov.reportPaths: <empty>
16:46:04.788 DEBUG 'Import of stylelint issues' skipped because of missing configuration requirements.
Accessed configuration:
- sonar.typescript.tslint.reportPaths: <empty>
- sonar.javascript.lcov.reportPaths: <empty>
- sonar.css.stylelint.reportPaths: <empty>
```

```
cs -> ThymeLeaf template sensor -> JavaAndroidConfigurationSensor -> SurefireSensor -> HTML -> Declarative Rule Engine for Shell -> X
ML Sensor -> JasminFileCollectorSensor -> Serverless configuration file sensor -> AWS SAM template file sensor -> AWS SAM Inline temp
late file sensor -> DBD/Java IR Generation -> javabugs -> pythonbugs -> EnterpriseTextAndSecretsSensor -> JavaSecuritySensor -> CShar
pSecuritySensor -> VbNetSecuritySensor -> PhpSecuritySensor -> PythonSecuritySensor -> JsSecuritySensor -> KotlinSecuritySensor -> Go
SecuritySensor
16:46:05.154 INFO  Sensor JavaSensor [java]
16:46:05.181 DEBUG Property 'sonar.java.jdkHome' resolved with:
[]
16:46:05.182 DEBUG Property 'sonar.java.libraries' resolved with:
[]
16:46:05.182 DEBUG Property 'sonar.java.jdkHome' resolved with:
[]
16:46:05.183 DEBUG Property 'sonar.java.test.libraries' resolved with:
[]
16:46:05.229 DEBUG Found 63 JSP files.
16:46:05.238 DEBUG Context root set to C:\Users\Hasib\Desktop\dvja\src\main\webapp
16:46:05.660 DEBUG Warning: No org.apache.tomcat.JarScanner set in ServletContext. Falling back to default JarScanner implementation.
16:46:05.837 ERROR [stderr] Oct 15, 2025 4:46:05 PM org.apache.jasper.servlet.TldScanner scanJars
16:46:05.838 ERROR [stderr] INFO: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger f
or a complete list of JARs that were scanned but no TLDs were found in them. Skipping unneeded JARs during scanning can improve start
up time and JSP compilation time.
16:46:05.844 DEBUG Transpiling JSP: C:\Users\Hasib\Desktop\dvja\src\main\webapp\WEB-INF\dvja\a6_sensitive_data\scenario.jsp
16:46:06.146 DEBUG Error transpiling src/main/webapp/WEB-INF/dvja/a6_sensitive_data/scenario.jsp. Error:
org.apache.jasper.JasperException: Unable to get JAR resource [/struts-tags] containing TLD: [java.io.FileNotFoundException]
        at org.apache.jasper.compiler.DefaultErrorHandler.jspError(DefaultErrorHandler.java:37)
        at org.apache.jasper.compiler.ErrorDispatcher.dispatch(ErrorDispatcher.java:270)
        at org.apache.jasper.compiler.ErrorDispatcher.jspError(ErrorDispatcher.java:77)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.generateTldResourcePath(TagLibraryInfoImpl.java:279)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.<init>(TagLibraryInfoImpl.java:122)
        at org.apache.jasper.compiler.Parser.parseTaglibDirective(Parser.java:415)
        at org.apache.jasper.compiler.Parser.parseDirective(Parser.java:468)
        at org.apache.jasper.compiler.Parser.parseElements(Parser.java:1361)
        at org.apache.jasper.compiler.Parser.parse(Parser.java:137)
```

```
        at org.sonar.scanner.bootstrap.ScannerMain.main(ScannerMain.java:42)

16:46:06.167 DEBUG Transpiling JSP: C:\Users\Hasib\Desktop\dvja\src\main\webapp\WEB-INF\dvja\a6_sensitive_data.jsp
16:46:06.190 DEBUG Error transpiling src/main/webapp/WEB-INF/dvja/a6_sensitive_data.jsp. Error:
org.apache.jasper.JasperException: Unable to get JAR resource [/struts-tags] containing TLD: [java.io.FileNotFoundException]
        at org.apache.jasper.compiler.DefaultErrorHandler.jspError(DefaultErrorHandler.java:37)
        at org.apache.jasper.compiler.ErrorDispatcher.dispatch(ErrorDispatcher.java:270)
        at org.apache.jasper.compiler.ErrorDispatcher.jspError(ErrorDispatcher.java:77)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.generateTldResourcePath(TagLibraryInfoImpl.java:279)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.<init>(TagLibraryInfoImpl.java:122)
        at org.apache.jasper.compiler.Parser.parseTaglibDirective(Parser.java:415)
        at org.apache.jasper.compiler.Parser.parseDirective(Parser.java:468)
        at org.apache.jasper.compiler.Parser.parseElements(Parser.java:1361)
        at org.apache.jasper.compiler.Parser.parse(Parser.java:137)
        at org.apache.jasper.compiler.ParserController.doParse(ParserController.java:243)
        at org.apache.jasper.compiler.ParserController.parse(ParserController.java:106)
        at org.apache.jasper.compiler.Compiler.generateJava(Compiler.java:193)
        at org.apache.jasper.compiler.Compiler.compile(Compiler.java:366)
        at org.sonar.java.jsp.Jasper.transpileJsp(Jasper.java:163)
        at org.sonar.java.jsp.Jasper.generateFiles(Jasper.java:100)
        at org.sonar.plugins.java.JavaSensor.runJasper(JavaSensor.java:169)
        at org.sonar.plugins.java.JavaSensor.execute(JavaSensor.java:128)
        at org.sonar.scanner.sensor.AbstractSensorWrapper.analyse(AbstractSensorWrapper.java:68)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.execute(ModuleSensorsExecutor.java:75)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.lambda$execute$1(ModuleSensorsExecutor.java:48)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.withModuleStrategy(ModuleSensorsExecutor.java:66)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.execute(ModuleSensorsExecutor.java:48)
        at org.sonar.scanner.scan.ModuleScanContainer.doAfterStart(ModuleScanContainer.java:64)
        at org.sonar.core.platform.ComponentContainer.startComponents(ComponentContainer.java:123)
        at org.sonar.core.platform.ComponentContainer.execute(ComponentContainer.java:109)
        at org.sonar.scanner.scan.ProjectScanContainer.scan(ProjectScanContainer.java:211)
        at org.sonar.scanner.scan.ProjectScanContainer.scanRecursively(ProjectScanContainer.java:207)
        at org.sonar.scanner.scan.ProjectScanContainer.doAfterStart(ProjectScanContainer.java:170)
```

```
16:46:06.524 DEBUG Transpiling JSP: C:\Users\Hasib\Desktop\dvja\src\main\webapp\WEB-INF\dvja\a3_xss\scenario.jsp
16:46:06.533 DEBUG Error transpiling src/main/webapp/WEB-INF/dvja/a3_xss/scenario.jsp. Error:
org.apache.jasper.JasperException: Unable to get JAR resource [/struts-tags] containing TLD: [java.io.FileNotFoundException]
        at org.apache.jasper.compiler.DefaultErrorHandler.jspError(DefaultErrorHandler.java:37)
        at org.apache.jasper.compiler.ErrorDispatcher.dispatch(ErrorDispatcher.java:270)
        at org.apache.jasper.compiler.ErrorDispatcher.jspError(ErrorDispatcher.java:77)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.generateTldResourcePath(TagLibraryInfoImpl.java:279)
        at org.apache.jasper.compiler.TagLibraryInfoImpl.<init>(TagLibraryInfoImpl.java:122)
        at org.apache.jasper.compiler.Parser.parseTaglibDirective(Parser.java:415)
        at org.apache.jasper.compiler.Parser.parseDirective(Parser.java:468)
        at org.apache.jasper.compiler.Parser.parseElements(Parser.java:1361)
        at org.apache.jasper.compiler.Parser.parse(Parser.java:137)
        at org.apache.jasper.compiler.ParserController.doParse(ParserController.java:243)
        at org.apache.jasper.compiler.ParserController.parse(ParserController.java:106)
        at org.apache.jasper.compiler.Compiler.generateJava(Compiler.java:193)
        at org.apache.jasper.compiler.Compiler.compile(Compiler.java:366)
        at org.sonar.java.jsp.Jasper.transpileJsp(Jasper.java:163)
        at org.sonar.java.jsp.Jasper.generateFiles(Jasper.java:100)
        at org.sonar.plugins.java.JavaSensor.runJasper(JavaSensor.java:169)
        at org.sonar.plugins.java.JavaSensor.execute(JavaSensor.java:128)
        at org.sonar.scanner.sensor.AbstractSensorWrapper.analyse(AbstractSensorWrapper.java:68)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.execute(ModuleSensorsExecutor.java:75)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.lambda$execute$1(ModuleSensorsExecutor.java:48)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.withModuleStrategy(ModuleSensorsExecutor.java:66)
        at org.sonar.scanner.sensor.ModuleSensorsExecutor.execute(ModuleSensorsExecutor.java:48)
        at org.sonar.scanner.scan.ModuleScanContainer.doAfterStart(ModuleScanContainer.java:64)
        at org.sonar.core.platform.ComponentContainer.startComponents(ComponentContainer.java:123)
        at org.sonar.core.platform.ComponentContainer.execute(ComponentContainer.java:109)
        at org.sonar.scanner.scan.ProjectScanContainer.scan(ProjectScanContainer.java:211)
        at org.sonar.scanner.scan.ProjectScanContainer.scanRecursively(ProjectScanContainer.java:207)
        at org.sonar.scanner.scan.ProjectScanContainer.doAfterStart(ProjectScanContainer.java:170)
        at org.sonar.core.platform.ComponentContainer.startComponents(ComponentContainer.java:123)
```

```
16:46:07.221 DEBUG 'src/main/java/com/appsecco/dvja/controllers/Home.java' generated metadata with charset 'windows-1252'
16:46:07.224 DEBUG 'src/main/java/com/appsecco/dvja/services/ProductService.java' generated metadata with charset 'windows-1252'
16:46:07.229 DEBUG 'src/main/java/com/appsecco/dvja/services/UserRegistrationService.java' generated metadata with charset 'windows-1
252'
16:46:07.232 DEBUG 'src/main/java/com/appsecco/dvja/models/User.java' generated metadata with charset 'windows-1252'
16:46:07.234 DEBUG 'src/main/java/com/appsecco/dvja/interceptors/AuthenticationInterceptor.java' generated metadata with charset 'win
dows-1252'
16:46:07.235 DEBUG 'src/main/java/com/appsecco/dvja/controllers/ResetPassword.java' generated metadata with charset 'windows-1252'
16:46:07.239 DEBUG 'src/main/java/com/appsecco/dvja/controllers/ProductAction.java' generated metadata with charset 'windows-1252'
16:46:07.241 DEBUG 'src/main/java/com/appsecco/dvja/models/Product.java' generated metadata with charset 'windows-1252'
16:46:07.246 DEBUG 'src/main/java/com/appsecco/example/HelloWorld.java' generated metadata with charset 'windows-1252'
16:46:07.251 DEBUG 'src/main/java/com/appsecco/dvja/services/UserService.java' generated metadata with charset 'windows-1252'
16:46:07.253 DEBUG 'src/main/java/com/appsecco/example/ExampleSupport.java' generated metadata with charset 'windows-1252'
16:46:08.321 INFO  Starting batch processing.
16:46:09.817 INFO  The Java analyzer cannot skip unchanged files in this context. A full analysis is performed for all files.
16:46:10.016 DEBUG Generating IR for file:///C:/Users/Hasib/Desktop/dvja/src/main/java/com/appsecco/dvja/controllers/ApiAction.java
16:46:10.368 DEBUG Encountered impossible type that is not primitive, void, array, interface or class: !Unknown!
16:46:10.384 DEBUG Found no nullary method named iterator in type !unknown!
16:46:10.832 DEBUG Incomplete Semantic, method invocation 'equals' line 40 col 60
16:46:10.907 DEBUG Incomplete Semantic, method invocation 'isEmpty' line 70 col 24
16:46:11.865 DEBUG Incomplete Semantic, method invocation 'equals' line 40 col 60
16:46:11.903 DEBUG Incomplete Semantic, method invocation 'equals' line 40 col 60
16:46:11.909 DEBUG Incomplete Semantic, method invocation 'isEmpty' line 70 col 24
16:46:12.336 DEBUG Analysis time of src/main/java/com/appsecco/dvja/controllers/ApiAction.java (2758ms)
16:46:12.427 DEBUG Generating IR for file:///C:/Users/Hasib/Desktop/dvja/src/main/java/com/appsecco/dvja/services/UserAuthenticationS
ervice.java
16:46:12.861 DEBUG [SE] Loaded 255 hardcoded method behaviors.
16:46:13.008 DEBUG Generating IR for file:///C:/Users/Hasib/Desktop/dvja/src/main/java/com/appsecco/example/Login.java
16:46:13.021 DEBUG Encountered impossible type that is not primitive, void, array, interface or class: !Unknown!
16:46:13.034 DEBUG Incomplete Semantic, unknown identifier 'SUCCESS' line 32 col 16
16:46:13.111 DEBUG Incomplete Semantic, unknown identifier 'SUCCESS' line 32 col 16
16:46:13.254 DEBUG Generating IR for file:///C:/Users/Hasib/Desktop/dvja/src/main/java/com/appsecco/dvja/controllers/PingAction.java
16:46:13.265 DEBUG Encountered impossible type that is not primitive, void, array, interface or class: !Unknown!
```

```
y. Verify your configuration, as you might end up with less precise results.
16:46:20.299 WARN  Unresolved imports/types have been detected during analysis. Enable DEBUG mode to see them.
16:46:20.313 DEBUG Unresolved imports/types:
- ActionInvocation cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/interceptors/AuthenticationInterceptor.java

- ActionSupport cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/controllers/BaseController.java
  * src/main/java/com/appsecco/example/ExampleSupport.java

- Column cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/models/User.java
  * src/main/java/com/appsecco/dvja/models/Product.java

- Cookie cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/controllers/ApiAction.java

- Entity cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/models/User.java
  * src/main/java/com/appsecco/dvja/models/Product.java

- EntityManager cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/services/ProductService.java
  * src/main/java/com/appsecco/dvja/services/UserService.java

- GeneratedValue cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/models/User.java
  * src/main/java/com/appsecco/dvja/models/Product.java

- Gson cannot be resolved to a type
  * src/main/java/com/appsecco/dvja/controllers/BaseController.java

- GsonBuilder cannot be resolved to a type
```

```
S3330,S6842,S6843,S2123,S3696,S4423,S2486,S6840,S2245,S6841,S6846,S2004,S3699,S5876,S6847,S3579,S6844,S4426,S5757,S6845,S5759,S6848,S
6035,S5860,S6275,S6397,S2234,S3686,S5863,S2598,S6836,S6957,S1940,S5868,S5869,S6958,S6959,S1135,S1134,S6281,S6265,S5730,S5850,S6821,S5
732,S6268,S3796,S5852,S5973,S4524,S5734,S6824,S125,S6825,S128,S3799,S2589,S5736,S6822,S5856,S6823,S5739,S6827,S6270,S1264,S1143,S1481
,S2692,S3782,S5842,S3785,S4634,S5843,S5725,S5604,S6019,S2699,S6811,S5728,S1119,S6819,S4084,S6486,S2681,S1472,S6249,S3531,S2685,S2201,
S4502,S3776,S2688,S1479,S4624,S3415,S5958,S6807,S4507,S1128,S1126,S1125,S1121,S6252,S5042,S5264,S6594,S6353,S6479,S2430,S2310,S6477,S
2432,S5148,S6478,S3403,S1219,S4619,S6481,S4043,S4165,S5254,S5257,S5256,S2301,S4721,S3516,S1226,S5260,S6351,S6332,S7784,S6333,S7785,S6
330,S5122,S4275,S6331,S7783,S4036,S3981,S4830,S7786,S4158,S5247,S7787,S3500,S3984,S3863,S3504,S3626,S1314,S1313,S6582,S6442,S6321,S77
73,S6443,S7774,S4144,S6440,S7771,S4143,S6441,S7772,S6325,S7777,S6326,S7778,S6323,S3972,S6324,S7776,S6329,S3854,S6327,S4822,S6328,S373
5,S3616,S1439,S1321,S7780,S2092,S7781,S4030,S2094,S7762,S7763,S6671,S7760,S6551,S6793,S7761,S878,S2990,S2077,S6435,S7766,S2870,S6557,
S7767,S2871,S7764,S6676,S7765,S6439,S4138,S6319,S6679,S7768,S2755,S6438,S6317,S7769,S1788,S2999,S2757,S2639,S7770,S4140,S5693,S7751,S
7752,S5332,S6660,S6661,S7750,S6303,S6666,S2187,S7755,S888,S4123,S6788,S7756,S2189,S2068,S7753,S4125,S6302,S6544,S7754,S7759,S6308,S15
33,S6426,S6789,S7757,S1534,S7758,S1656,S3834,S1536,S1301,S6790,S6791,S6772,S1082,S7740,S7741,S6770,S6650,S6534,S7744,S5443,S6535,S774
5,S2970,S6653,S6774,S7742,S6654,S6775,S7743,S7748,S1763,S5689,S6418,S7749,S1764,S6657,S7746,S1523,S2612,S7747,S1527,S2737,S1529,S1199
,S1077,S5691,S6761,S7059,S905,S1090,S108,S6523,S6644,S7733,S107,S6645,S6766,S7734,S6763,S1871,S6522,S6643,S7732,S1751,S7737,S1994,S77
38,S1874,S6767,S7735,S6647,S7736,S3812,S1515,S7739,S1516,S101,S6092,S7060,S2392,S6079,S6750,S3001,S6754,S7722,S5542,S7723,S7720,S7721
,S5547,S6637,S7726,S1862,S6638,S6635,S6756,S7724,S3800,S6757,S7725,S3923,S7728,S7729,S6080,S6861,S6747,S3358,S6748,S6627,S5659,S6746,
S1854,S6509,S7719,S2703,S6749,S7718,S1068,S1186,S5869,S6958,S6959,S2598,S6836,S6957,S5868,S2234,S5863,S1143,S1264,S5860,S5759,S6606,S
6848,S6846,S2004,S3699,S5876,S6847,S3579,S6844,S4426,S5757,S6845,S6842,S6843,S3696,S2123,S4423,S2486,S6840,S2245,S6841,S4782,S3330,S6
859,S2137,S5527,S6853,S6851,S6852,S4790,S6850,S2251,S7719,S6749,S6509,S7718,S6747,S4325,S3358,S6748,S5659,S6627,S6746,S4322,S4323,S68
61,S4619,S1226,S2430,S2310,S1479,S5958,S6807,S4507,S4623,S4502,S3776,S1119,S2688,S4624,S3415,S4621,S3531,S2685,S2201,S2681,S1472,S681
9,S1125,S5728,S4634,S5843,S5725,S5604,S2699,S6811,S1128,S5842,S1121,S2692,S1135,S5739,S6827,S4524,S5734,S6824,S6825,S3799,S2589,S5736
,S6822,S5856,S6823,S5730,S5850,S6821,S5732,S5852,S5973,S1134,S2639,S2757,S2999,S1301,S1788,S2755,S2990,S2870,S2871,S3735,S1314,S1313,
S3616,S1439,S3854,S4822,S3972,S3504,S3626,S1444,S3984,S3863,S3981,S4830,S3516,S1219,S2301,S4721,S3923,S1862,S6079,S3812,S1874,S1994,S
1751,S1516,S1515,S1871,S7059,S6080,S1529,S1523,S1764,S1763,S2737,S2612,S2970,S6092,S7060,S1534,S1533,S1656,S905,S125,S128,S1940,S6397
,S6275,S6035,S6270,S6281,S2819,S1607,S1848,S2933,S1854,S6478,S6477,S6598,S5148,S6479,S6353,S5264,S6594,S5260,S6590,S6351,S6249,S6486,
S6481,S6019,S5042,S6252,S6268,S6265,S4084,S6438,S6317,S7649,S4138,S6319,S6679,S7647,S7768,S6439,S7648,S7769,S2077,S7766,S7767,S6557,S
7764,S6435,S6676,S7765,S6551,S7641,S7762,S6550,S7763,S6671,S7760,S6793,S7761,S878,S6790,S6791,S6328,S6327,S6569,S6329,S6324,S7656,S77
77,S6323,S6565,S7778,S6326,S6568,S7654,S6325,S7655,S7776,S6441,S7652,S7773,S6440,S7653,S7774,S4144,S6443,S6564,S7650,S7771,S4143,S644
2,S6321,S7651,S7772,S888,S7770,S4140,S4036,S4156,S7786,S4158,S5247,S6578,S7787,S6331,S7784,S6330,S6572,S7785,S6333,S4275,S5122,S6332,
S7783,S7780,S2092,S7781,S4030,S6571,S2094,S5257,S5256,S4043,S6583,S4165,S5254,S6582,S7728,S7729,S5547,S6637,S7726,S6638,S6759,S6635,S
6756,S7724,S6757,S7725,S3001,S6754,S7722,S5542,S7723,S4335,S7720,S7721,S2392,S1068,S6750,S1186,S7739,S7737,S7738,S6767,S7735,S6647,S7
736,S6522,S6644,S7733,S6766,S7734,S6763,S6523,S6643,S7732,S6761,S1199,S1077,S101,S7748,S5689,S7749,S6657,S6418,S7746,S7747,S7744,S544
3,S7745,S6535,S6653,S7742,S6534,S6654,S6775,S7743,S6772,S7740,S7741,S6770,S6650,S1082,S7759,S6426,S6308,S6789,S7757,S7758,S4124,S6302
```

```
16:46:53.937 DEBUG 'src/main/webapp/assets/fa/scss/_stacked.scss' generated metadata with charset 'windows-1252'
16:46:53.942 DEBUG Setting js/ts exclusions to C:/Users/Hasib/Desktop/dvja/**/.git/**,C:/Users/Hasib/Desktop/dvja/**/node_modules/**,
C:/Users/Hasib/Desktop/dvja/**/bower_components/**,C:/Users/Hasib/Desktop/dvja/**/dist/**,C:/Users/Hasib/Desktop/dvja/**/vendor/**,C:
/Users/Hasib/Desktop/dvja/**/external/**,C:/Users/Hasib/Desktop/dvja/**/contrib/**,C:/Users/Hasib/Desktop/dvja/**/*.d.ts,C:/Users/Has
ib/Desktop/dvja/.scannerwork
16:46:53.950 DEBUG Found 0 issue(s)
16:46:53.950 INFO  31/31 source files have been analyzed
16:46:53.951 INFO  Hit the cache for 0 out of 0
16:46:53.951 INFO  Miss the cache for 0 out of 0
16:46:53.951 INFO  Sensor CSS Rules [javascript] (done) | time=2526ms
16:46:53.951 INFO  Sensor CSS Metrics [javascript]
16:46:56.448 INFO  Sensor CSS Metrics [javascript] (done) | time=2497ms
16:46:56.449 INFO  Sensor ThymeLeaf template sensor [securityjavafrontend]
16:46:56.452 INFO  Sensor ThymeLeaf template sensor [securityjavafrontend] (done) | time=4ms
16:46:56.454 INFO  Sensor JavaAndroidConfigurationSensor [securityjavafrontend]
16:46:56.457 INFO  Sensor JavaAndroidConfigurationSensor [securityjavafrontend] (done) | time=4ms
16:46:56.458 INFO  Sensor SurefireSensor [java]
16:46:56.460 INFO  parsing [C:\Users\Hasib\Desktop\dvja\target\surefire-reports]
16:46:56.462 INFO  Sensor SurefireSensor [java] (done) | time=4ms
16:46:56.463 INFO  Sensor HTML [web]
16:46:56.472 DEBUG 'src/main/webapp/WEB-INF/dvja/a6_sensitive_data/scenario.jsp' generated metadata with charset 'windows-1252'
16:46:56.544 DEBUG Not enough content in 'src/main/webapp/WEB-INF/dvja/a6_sensitive_data/scenario.jsp' to have CPD blocks, it will no
t be part of the duplication detection
16:46:56.607 DEBUG 'src/main/webapp/WEB-INF/dvja/a2_broken_auth/scenario.jsp' generated metadata with charset 'windows-1252'
16:46:56.620 DEBUG Not enough content in 'src/main/webapp/WEB-INF/dvja/a2_broken_auth/scenario.jsp' to have CPD blocks, it will not b
e part of the duplication detection
16:46:56.634 DEBUG 'src/main/webapp/WEB-INF/dvja/a6_sensitive_data.jsp' generated metadata with charset 'windows-1252'
16:46:56.711 DEBUG Not enough content in 'src/main/webapp/WEB-INF/dvja/a5_sec_misconf/reference.jsp' to have CPD blocks, it will not
be part of the duplication detection
16:46:56.723 DEBUG 'src/main/webapp/WEB-INF/dvja/a9_vuln_component.jsp' generated metadata with charset 'windows-1252'
16:46:56.778 DEBUG 'src/main/webapp/WEB-INF/dvja/common/Menu.jsp' generated metadata with charset 'windows-1252'
16:46:56.817 DEBUG Not enough content in 'src/main/webapp/WEB-INF/dvja/a9_vuln_component/description.jsp' to have CPD blocks, it will
 not be part of the duplication detection
```

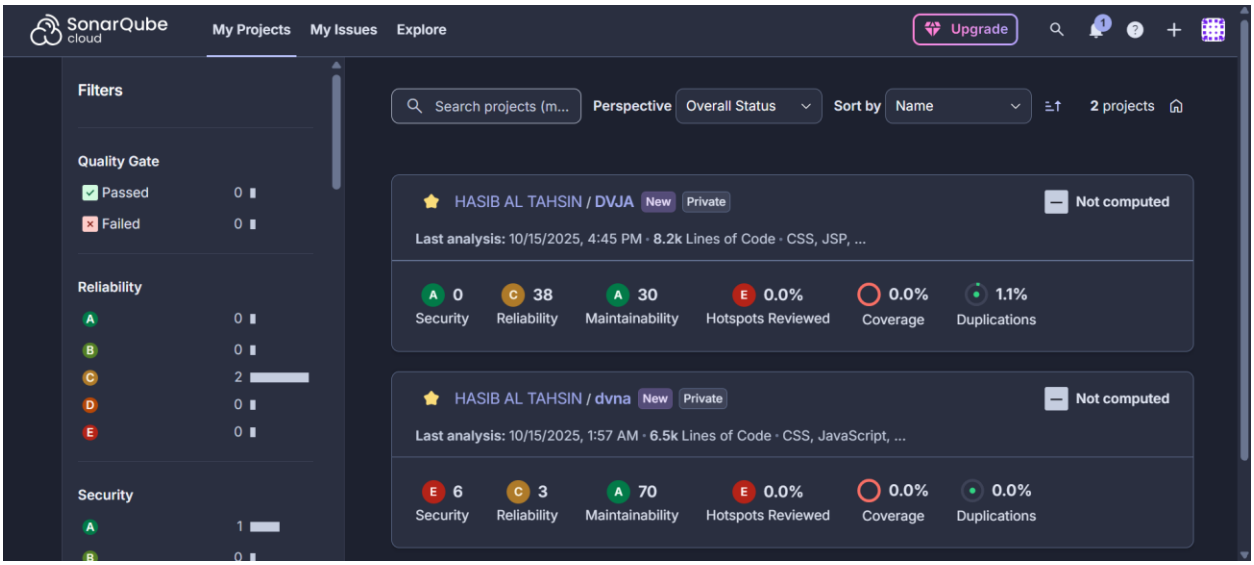
```
16:47:45.635 DEBUG <-- 200 OK https://analysis-sensorcache-eu-central-1-prod.s3.eu-central-1.amazonaws.com/AZnj8JSpXQjBuNsvoFuz/AZnnc
0bOwGJ01I0ac6JI/AZnnc0bOwGJ01I0ac6JI?X-Amz-Security-Token=IQoJb3JpZ2luX2VjEMn%2F%2F%2F%2F%2F%2F%2F%2F%2F%2FwEaDGV1LWNlbnRyYWwtMSJGMEQ
CIALbI4oQv1yIllxiMrY5lrCCjOKT3PiRYga9VMYxNm0cAiACnyp2mTSSam8glz01k3%2FP5nMU4pbHihcFoY5qhp8eySqmAwhyEAMaDDQ4ODA1OTk2NTYzNSIMhU6YhQwbXx
vE4geUkoMDtT%2BU75dIDAk%2Fwde26HzoKpwmCPb8AziKOL0UOEu0gdObT35yVSZn7wT%2B1Eh55x5QGrB3OZuVA%2FmRGkRetBy08CkNkKeYp%2FnzaecqHuRF7c26dTYuz
k%2FlVViKAmPtgh9HMYH4wM2qVhiM%2FMnEPEH13cZQ89%2FuY3xx8N5CyVhVviv%2FU7RMC4m3QPlTvm0n7uYNlf3ozvg1TSBlaE6a9qG67IFg2qCDzsoaFQiQaV1Wj0qFzJ
0MUzYov4Gr6v1uFqVwnBX7YnNFXoGuVf4wogMzTaGkfMAuTALi9SDVmyXiAemfLit%2B%2BqdyqcCymli51BkGTNXLKHqarjAveMpT4LlvuM14%2F1JyPQTz6hJXIdT1APLgY
k9k5Go6QYitqPXPwtcw8MqKKOxeNOuT4bPTmTZiqObLfrNWRBd%2FRvY9ABMAGP%2BvwYehA%2FpoBU0tW90D990MW8D%2FETPjqg3qJYyxAL7BqJkB62%2Fr1uXpueaz6%2B
sDyvli7sv%2BqgWX80Pp4ehVls9%2F1siaMOS9vccGOp4BXm9ndTYBrLNfYY4J0w%2FrMYAtNTjN7rVmkG7%2FVFSlQZHwUdYafhn6khKGvWv3TMqoe3X4D3l2Eferjor5S8A
%2BFbfxYYLkEH4p0Ht9e0MR6lY009MCuKj9c5HIrPQfCsAPlJGfLOvWafZLV4LitCRaSybbrqYyaOvP0qERM24rxLLEXVvwOy5vC1KwhvJXvz%2FQfSTRoY%2BdQs7FjMWU%2
FB4%3D&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20251015T104744Z&X-Amz-SignedHeaders=host&X-Amz-Credential=ASIAXDIVECDBRGYJZPG4%2F
20251015%2Feu-central-1%2Fs3%2Faws4_request&X-Amz-Expires=900&X-Amz-Signature=e035534193084d9f696d5a03f719eec26a6877d7e61f617dfd30046
007d8a0df (1146ms, 0-byte body)
16:47:45.636 INFO  Sensor cache published successfully
16:47:45.638 DEBUG Architecture analysis is disabled
16:47:45.641 DEBUG Post-jobs :
16:47:45.648 DEBUG Shutting down the worker
16:47:45.651 INFO  Rule | Time (ms) | Relative
16:47:45.652 INFO  :----|----------:|--------:
16:47:45.706 DEBUG The worker thread exited with code 0
16:47:45.706 DEBUG Closing server
16:47:45.707 DEBUG WebSocket connection closed:  (code: 1006)
16:47:45.711 DEBUG WebSocket client disconnected:  with code 1006
16:47:45.711 DEBUG Closed WebSocket connection
16:47:45.714 DEBUG The bridge server shut down
16:47:45.715 INFO  Rule | Time (ms) | Relative
16:47:45.716 INFO  :----|----------:|--------:
16:47:46.222 INFO  Analysis total time: 1:57.307 s
16:47:46.226 INFO  SonarScanner Engine completed successfully
16:47:46.257 DEBUG Cleanup org.eclipse.jgit.util.FS$FileStoreAttributes$$Lambda$268/0x00000222812eb958@34819867 during JVM shutdown
16:47:46.823 INFO  EXECUTION SUCCESS
16:47:46.826 INFO  Total time: 2:08.997s
```

The execution was found successful.

- -X debug mode, showed all information.
- The scanner .java files were analyzed, because the compiled classes path was correct.

Viewing SonarCloud Reports (GUI)

First, I opened SonarCloud in the browser. Then I went to my Organization and clicked on Projects - HASIB AL TAHSIN-DVJA. Immediately after that, various analysis indicators were displayed in the User Interface (UI) of SonarCloud, such as - Security, Reliability, Maintainability, Duplications, and Coverage etc.



For DVNA:

By using snyk code test in dvna:

```
C:\Users\Hasib\sast-lab\dvna>snyk code test

Testing C:\Users\Hasib\sast-lab\dvna ...

Open Issues

 ✗ [LOW] Sensitive Cookie with 'Secure' Attribute Set to False
   Finding ID: e99e5d47-449c-4c88-94e4-f8c67382103e
   Path: server.js, line 27
   Info: Cookie has the Secure attribute set to false. Set it to true to protect the cookie from man-in-the-middle atta
ks.

 ✗ [LOW] Use of Password Hash With Insufficient Computational Effort
   Finding ID: 42c9da9f-65b8-49cd-9de2-c7d452d45811
   Path: core/authHandler.js, line 78
   Info: MD5 hash (used in md5) is insecure. Consider changing it to a secure hashing algorithm.

 ✗ [LOW] Use of Password Hash With Insufficient Computational Effort
   Finding ID: eafb8198-c87c-46d0-b1b6-98ecdc12e58a
   Path: core/authHandler.js, line 49
   Info: MD5 hash (used in md5) is insecure. Consider changing it to a secure hashing algorithm.

 ✗ [LOW] Improper Type Validation
   Finding ID: b7d29545-2faf-40f5-bfee-047a1c13e16e
   Path: core/appHandler.js, line 151
```

```
uct the HTML page on client side. This may result in a DOM Based Cross-Site Scripting attack (DOMXSS).

 ✗ [MEDIUM] Allocation of Resources Without Limits for Login Operations
   Finding ID: 07549f31-05b0-4f2b-8024-50f39c7fce5e
   Path: routes/main.js, line 51
   Info: Call to authenticate performs user authentication, but is not protected by any rate-limiting mechanism. This ma
y enable the attackers to bruteforce users' passwords. Consider using a rate-limiting middleware such as express-limit.

 ✗ [MEDIUM] DOM-based Cross-site Scripting (XSS)
   Finding ID: e40d4a77-83e8-4640-b4f8-1ddefdd84fd3
   Path: views/app/adminusers.ejs, line 41
   Info: Unsanitized input from data from a remote resource flows into innerHTML, where it is used to dynamically constr
uct the HTML page on client side. This may result in a DOM Based Cross-Site Scripting attack (DOMXSS).

 ✗ [MEDIUM] DOM-based Cross-site Scripting (XSS)
   Finding ID: 3e612f19-1d07-4f64-890a-c949bb77e5a6
   Path: views/app/adminusers.ejs, line 40
   Info: Unsanitized input from data from a remote resource flows into innerHTML, where it is used to dynamically constr
uct the HTML page on client side. This may result in a DOM Based Cross-Site Scripting attack (DOMXSS).

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: f4b0d0b7-87b5-463c-bedc-7a7034d96302
   Path: core/appHandler.js, line 38
   Info: Expensive operation (a system command execution) is performed by an endpoint handler which does not use a rate-
limiting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting mid
dleware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: ce7318ec-2674-4b04-8700-7201975d01d2
   Path: core/appHandler.js, line 76
```

```
 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: 6696efe4-36d2-413e-8487-7247e4e757f2
   Path: core/appHandler.js, line 233
   Info: Expensive operation (a file system operation) is performed by an endpoint handler which does not use a rate-lim
iting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting middle
ware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: 490e05d2-16ce-4b34-94dc-fa321b375f37
   Path: core/authHandler.js, line 69
   Info: Expensive operation (a file system operation) is performed by an endpoint handler which does not use a rate-lim
iting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting middle
ware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: df40e62e-bf81-4fd9-aad1-3dc0cebdc43e
   Path: routes/app.js, line 10
   Info: Expensive operation (a file system operation) is performed by an endpoint handler which does not use a rate-lim
iting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting middle
ware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: 23770baf-da00-4321-a608-5845578b1a2d
   Path: routes/app.js, line 16
   Info: Expensive operation (a file system operation) is performed by an endpoint handler which does not use a rate-lim
iting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting middle
ware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
```

```
ware such as express-limit.

 ✗ [MEDIUM] Allocation of Resources Without Limits or Throttling
   Finding ID: 810dfff3-5c5c-47ff-b837-71a9048097da
   Path: routes/main.js, line 32
   Info: Expensive operation (a file system operation) is performed by an endpoint handler which does not use a rate-lim
iting mechanism. It may enable the attackers to perform Denial-of-service attacks. Consider using a rate-limiting middle
ware such as express-limit.

 ✗ [HIGH] Hardcoded Non-Cryptographic Secret
   Finding ID: cd17009f-a011-485c-a936-093318bc0823
   Path: server.js, line 23
   Info: Avoid hardcoding values that are meant to be secret. Found a hardcoded string used in express-session.

 ✗ [HIGH] Command Injection
   Finding ID: e90df790-9a4e-4ff7-b56b-4493fa5cc88a
   Path: core/appHandler.js, line 39
   Info: Unsanitized input from the HTTP request body flows into child_process.exec, where it is used to build a shell c
ommand. This may result in a Command Injection vulnerability.

 ✗ [HIGH] SQL Injection
   Finding ID: 7ea19b63-8826-4203-84d6-a5c12487ca75
   Path: core/appHandler.js, line 11
   Info: Unsanitized input from the HTTP request body flows into query, where it is used in an SQL query. This may resul
t in an SQL Injection vulnerability.

 ✗ [HIGH] Deserialization of Untrusted Data
   Finding ID: a3ffdecc-d41c-4672-b4f9-50e41628f875
   Path: core/appHandler.js, line 218
   Info: Unsanitized input from an uploaded file flows into node-serialize.unserialize, where it is used to deserialize
```

```
   Finding ID: 7ea19b63-8826-4203-84d6-a5c12487ca75
   Path: core/appHandler.js, line 11
   Info: Unsanitized input from the HTTP request body flows into query, where it is used in an SQL query. This may resul
t in an SQL Injection vulnerability.

 ✗ [HIGH] Deserialization of Untrusted Data
   Finding ID: a3ffdecc-d41c-4672-b4f9-50e41628f875
   Path: core/appHandler.js, line 218
   Info: Unsanitized input from an uploaded file flows into node-serialize.unserialize, where it is used to deserialize
an object. This may result in an Unsafe Deserialization vulnerability.



 Test Summary

   Organization:      hasibaltahsin
   Test type:         Static code analysis
   Project path:      C:\Users\Hasib\sast-lab\dvna

   Total issues:   36
   Ignored issues: 0 [ 0 HIGH  0 MEDIUM  0 LOW ]
   Open issues:    36 [ 4 HIGH  27 MEDIUM  5 LOW ]

💡 Tip

   To view ignored issues, use the --include-ignores option.
```

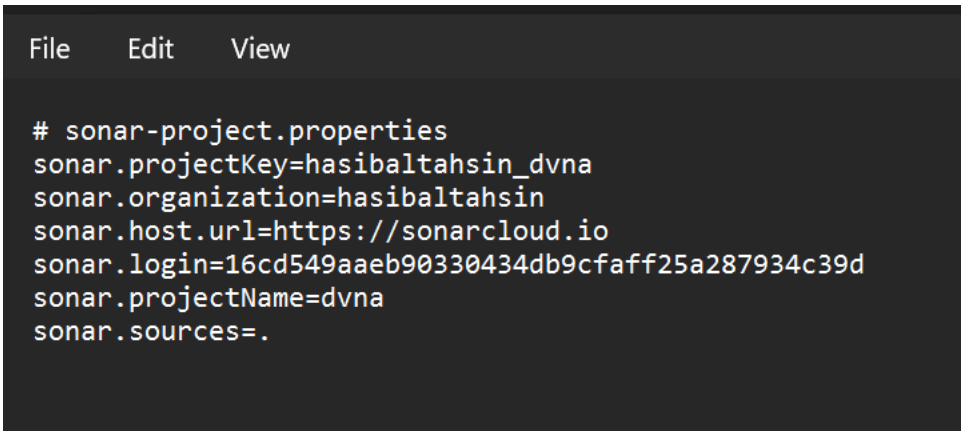
Set JAVA_HOME to my JDK 25 installation & Prepend Java bin to PATH:

```
PS C:\Users\Hasib\sast-lab\dvna> #1  Set JAVA_HOME to your JDK 25 installation
PS C:\Users\Hasib\sast-lab\dvna> $env:JAVA_HOME = "C:\Program Files\Eclipse Adoptium\jdk-25.0.0.36-hotspot"
PS C:\Users\Hasib\sast-lab\dvna>
PS C:\Users\Hasib\sast-lab\dvna> #2  Prepend Java bin to PATH
PS C:\Users\Hasib\sast-lab\dvna> $env:PATH = "$env:JAVA_HOME\bin;" + $env:PATH
PS C:\Users\Hasib\sast-lab\dvna>
PS C:\Users\Hasib\sast-lab\dvna> #3  Verify Java version
PS C:\Users\Hasib\sast-lab\dvna> java -version
openjdk version "25" 2025-09-16 LTS
OpenJDK Runtime Environment Temurin-25+36 (build 25+36-LTS)
OpenJDK 64-Bit Server VM Temurin-25+36 (build 25+36-LTS, mixed mode, sharing)
```

sonar-project.properties:

The following basic settings are stored in C:\Users\Hasib\sast-lab\dvna\sonar-project.properties:

```
PS C:\Users\Hasib\sast-lab\dvna> notepad "C:\Users\Hasib\sast-lab\dvna\sonar-project.properties"
```



View current directory list

The command I gave for this was:

Get-ChildItem (short: gci) PowerShell command. It showed the directories (folders) that belong to a specific folder. The log showed config, core, docs, models, node_modules, public, routes, views etc. This mean that my project structure was correct.

```
PS C:\Users\Hasib\sast-lab\dvna> Get-ChildItem "C:\Users\Hasib\sast-lab\dvna" -Directory


    Directory: C:\Users\Hasib\sast-lab\dvna


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/14/2025   5:27 PM                config
d-----        10/14/2025   5:27 PM                core
d-----        10/14/2025   5:27 PM                docs
d-----        10/14/2025   5:27 PM                models
d-----        10/14/2025  11:17 PM                node_modules
d-----        10/14/2025   5:27 PM                public
d-----        10/14/2025   5:27 PM                routes
d-----        10/14/2025   5:27 PM                views
```

Opened the sonar-project.properties file

What I ran:

This file told SonarScanner what sources, what properties to analyze using. Here I opened Notepad to check/edit the configuration.

```
PS C:\Users\Hasib\sast-lab\dvna> notepad "C:\Users\Hasib\sast-lab\dvna\sonar-project.properties"
```

Running Sonar Scanner

"&" ran the mentioned .bat script in PowerShell. This ran SonarScanner CLI and analyzed my project C:\Users\Hasib\sast-lab\dvna and uploaded it to SonarCloud (here SonarCloud) server and generated a report.

```
PS C:\Users\Hasib\sast-lab\dvna> & "C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-windows-x64\bin\s
onar-scanner.bat"
01:57:14.432 INFO  Scanner configuration file: C:\Users\Hasib\Downloads\sonar-scanner-cli-7.3.0.5189-windows-x64\sonar-scanner-7.3.0.5189-window
s-x64\bin\..\conf\sonar-scanner.properties
01:57:14.446 INFO  Project root configuration file: C:\Users\Hasib\sast-lab\dvna\sonar-project.properties
01:57:14.514 INFO  SonarScanner CLI 7.3.0.5189
01:57:14.521 INFO  Windows 11 10.0 amd64
01:57:14.585 WARN  Both 'sonar.login' and 'sonar.token' (or the 'SONAR_TOKEN' env variable) are set, but only the latter will be used.
01:57:16.634 INFO  Communicating with SonarQube Cloud
01:57:16.637 INFO  JRE provisioning: os[windows], arch[amd64]
01:57:20.060 INFO  Starting SonarScanner Engine...
01:57:20.064 INFO  Java 17.0.11 Eclipse Adoptium (64-bit)
01:57:22.768 INFO  Load global settings
01:57:24.049 INFO  Load global settings (done) | time=1286ms
01:57:24.060 INFO  Server id: 1BD809FA-AWHW8ct9-T_TB3XqouNu
01:57:24.412 INFO  Loading required plugins
01:57:24.413 INFO  Load plugins index
01:57:24.674 INFO  Load plugins index (done) | time=261ms
01:57:24.676 INFO  Load/download plugins
01:57:24.812 INFO  Load/download plugins (done) | time=136ms
01:57:25.751 INFO  Load project settings for component key: 'hasibaltahsin_dvna'
01:57:26.440 INFO  Load project settings for component key: 'hasibaltahsin_dvna' (done) | time=688ms
01:57:26.455 INFO  Process project properties
01:57:26.488 INFO  Execute project builders
01:57:26.490 INFO  Execute project builders (done) | time=1ms
01:57:26.506 INFO  Project key: hasibaltahsin_dvna
01:57:26.507 INFO  Base dir: C:\Users\Hasib\sast-lab\dvna
01:57:26.508 INFO  Working dir: C:\Users\Hasib\sast-lab\dvna\.scannerwork
01:57:26.526 INFO  Load project branches
01:57:27.198 INFO  Load project branches (done) | time=672ms
01:57:27.203 INFO  Check ALM binding of project 'hasibaltahsin_dvna'
01:57:27.741 INFO  Detected project binding: NOT_BOUND
01:57:27.742 INFO  Check ALM binding of project 'hasibaltahsin_dvna' (done) | time=539ms
```

```
01:58:42.870 INFO  Sensor JavaScript/TypeScript analysis [javascript] (done) | time=34782ms
01:58:42.942 INFO  Sensor JavaScript inside YAML analysis [javascript]
01:58:42.953 INFO  No input files found for analysis
01:58:42.958 INFO  Hit the cache for 0 out of 0
01:58:42.960 INFO  Miss the cache for 0 out of 0
01:58:42.962 INFO  Sensor JavaScript inside YAML analysis [javascript] (done) | time=15ms
01:58:42.964 INFO  Sensor CSS Rules [javascript]
01:58:43.086 INFO  30 source files to be analyzed
01:58:44.899 INFO  30/30 source files have been analyzed
01:58:44.902 INFO  Hit the cache for 0 out of 0
01:58:44.903 INFO  Miss the cache for 0 out of 0
01:58:44.905 INFO  Sensor CSS Rules [javascript] (done) | time=2052ms
01:58:44.907 INFO  Sensor CSS Metrics [javascript]
01:58:45.974 INFO  Sensor CSS Metrics [javascript] (done) | time=1074ms
01:58:45.979 INFO  Sensor IaC YAML Sensor [iacenterprise]
01:58:45.981 INFO  Sensor IaC YAML Sensor [iacenterprise] (done) | time=2ms
01:58:45.985 INFO  Sensor IaC JSON Sensor [iacenterprise]
01:58:45.986 INFO  Sensor IaC JSON Sensor [iacenterprise] (done) | time=0ms
01:58:45.987 INFO  Sensor Serverless configuration file sensor [security]
01:58:45.988 INFO  0 Serverless function entries were found in the project
01:58:46.003 INFO  0 Serverless function handlers were kept as entrypoints
01:58:46.003 INFO  Sensor Serverless configuration file sensor [security] (done) | time=17ms
01:58:46.004 INFO  Sensor AWS SAM template file sensor [security]
01:58:46.012 INFO  Sensor AWS SAM template file sensor [security] (done) | time=6ms
01:58:46.017 INFO  Sensor AWS SAM Inline template file sensor [security]
01:58:46.019 INFO  Sensor AWS SAM Inline template file sensor [security] (done) | time=5ms
01:58:46.020 INFO  Sensor EnterpriseTextAndSecretsSensor [textenterprise]
01:58:46.095 INFO  Available processors: 8
01:58:46.105 INFO  Using 8 threads for analysis.
01:58:48.232 INFO  The property "sonar.tests" is not set. To improve the analysis accuracy, we categorize a file as a test file if any of the fo
llowing is true:
  * The filename starts with "test"
  * The filename contains "test." or "tests."
  * Any directory in the file path is named: "doc", "docs", "test" or "tests"
  * Any directory in the file path has a name ending in "test" or "tests"
```

```
01:58:49.381 INFO  Starting the text and secrets analysis
01:58:49.384 INFO  52 source files to be analyzed for the text and secrets analysis
01:58:51.434 INFO  52/52 source files have been analyzed for the text and secrets analysis
01:58:51.441 INFO  Start fetching files for the binary file analysis
01:58:51.442 INFO  There are no files to be analyzed for the binary file analysis
01:58:51.443 INFO  1 file is ignored because it is untracked by git
01:58:51.445 INFO  Sensor EnterpriseTextAndSecretsSensor [textenterprise] (done) | time=5437ms
01:58:51.446 INFO  Sensor JavaSecuritySensor [security]
01:58:51.463 INFO  29 taint analysis rules enabled.
01:58:51.468 INFO  No UCFGs have been included for analysis.
01:58:51.475 INFO  java security sensor: Begin: 2025-10-14T19:58:51.452387600Z, End: 2025-10-14T19:58:51.471391500Z, Duration: 00:00:00.019
  Load type hierarchy and UCFGs: Begin: 2025-10-14T19:58:51.464391900Z, End: 2025-10-14T19:58:51.468389700Z, Duration: 00:00:00.003
    Load type hierarchy: Begin: 2025-10-14T19:58:51.464391900Z, End: 2025-10-14T19:58:51.467388400Z, Duration: 00:00:00.002
    Load UCFGs: Begin: 2025-10-14T19:58:51.467388400Z, End: 2025-10-14T19:58:51.467388400Z, Duration: 00:00:00.000
01:58:51.476 INFO  java security sensor peak memory: 85 MB
01:58:51.477 INFO  Sensor JavaSecuritySensor [security] (done) | time=30ms
01:58:51.477 INFO  Sensor CSharpSecuritySensor [security]
01:58:51.478 INFO  27 taint analysis rules enabled.
01:58:51.479 INFO  No UCFGs have been included for analysis.
01:58:51.480 INFO  csharp security sensor: Begin: 2025-10-14T19:58:51.476391600Z, End: 2025-10-14T19:58:51.478390100Z, Duration: 00:00:00.001
  Load type hierarchy and UCFGs: Begin: 2025-10-14T19:58:51.477392700Z, End: 2025-10-14T19:58:51.477392700Z, Duration: 00:00:00.000
    Load type hierarchy: Begin: 2025-10-14T19:58:51.477392700Z, End: 2025-10-14T19:58:51.477392700Z, Duration: 00:00:00.000
    Load UCFGs: Begin: 2025-10-14T19:58:51.477392700Z, End: 2025-10-14T19:58:51.477392700Z, Duration: 00:00:00.000
01:58:51.481 INFO  csharp security sensor peak memory: 85 MB
01:58:51.481 INFO  Sensor CSharpSecuritySensor [security] (done) | time=5ms
01:58:51.482 INFO  Sensor VbNetSecuritySensor [security]
01:58:51.483 INFO  25 taint analysis rules enabled.
01:58:51.483 INFO  No UCFGs have been included for analysis.
01:58:51.485 INFO  vbnet security sensor: Begin: 2025-10-14T19:58:51.481390600Z, End: 2025-10-14T19:58:51.483390900Z, Duration: 00:00:00.002
  Load type hierarchy and UCFGs: Begin: 2025-10-14T19:58:51.481390600Z, End: 2025-10-14T19:58:51.482391300Z, Duration: 00:00:00.001
    Load type hierarchy: Begin: 2025-10-14T19:58:51.481390600Z, End: 2025-10-14T19:58:51.482391300Z, Duration: 00:00:00.001
    Load UCFGs: Begin: 2025-10-14T19:58:51.482391300Z, End: 2025-10-14T19:58:51.482391300Z, Duration: 00:00:00.000
01:58:51.485 INFO  vbnet security sensor peak memory: 85 MB
01:58:51.486 INFO  Sensor VbNetSecuritySensor [security] (done) | time=4ms
01:58:51.487 INFO  Sensor PhpSecuritySensor [security]
01:58:51.488 INFO  18 taint analysis rules enabled.
```

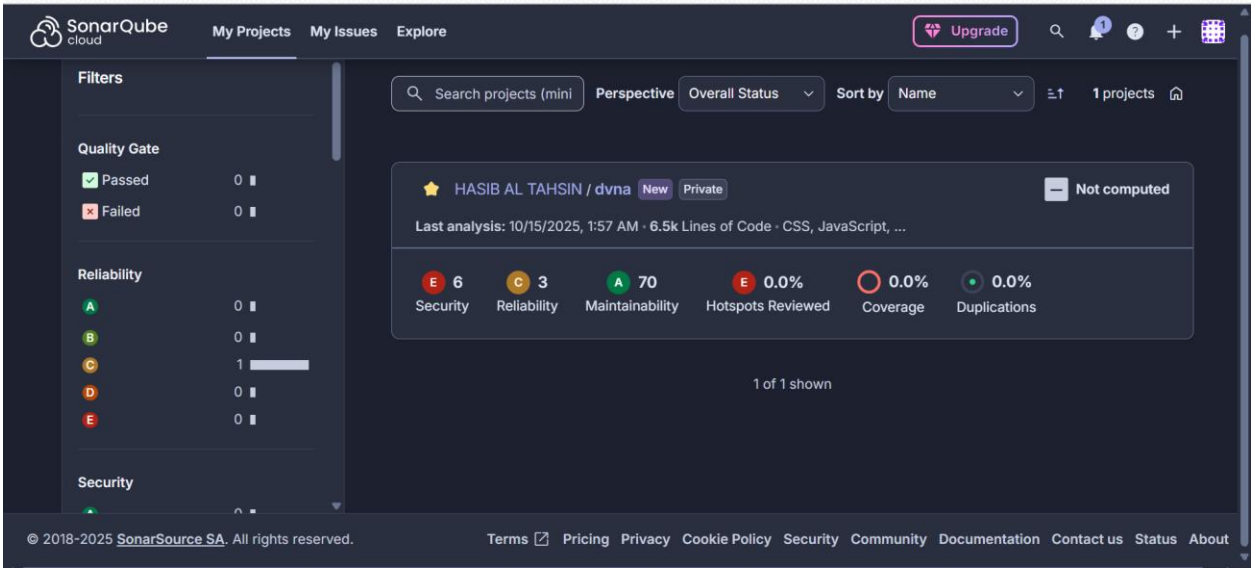
```
01:58:51.514 INFO  go security sensor: Begin: 2025-10-14T19:58:51.502390600Z, End: 2025-10-14T19:58:51.505392700Z, Duration: 00:00:00.003
  Load type hierarchy and UCFGs: Begin: 2025-10-14T19:58:51.502390600Z, End: 2025-10-14T19:58:51.503389500Z, Duration: 00:00:00.000
    Load type hierarchy: Begin: 2025-10-14T19:58:51.502390600Z, End: 2025-10-14T19:58:51.502390600Z, Duration: 00:00:00.000
    Load UCFGs: Begin: 2025-10-14T19:58:51.502390600Z, End: 2025-10-14T19:58:51.502390600Z, Duration: 00:00:00.000
01:58:51.515 INFO  go security sensor peak memory: 85 MB
01:58:51.515 INFO  Sensor GoSecuritySensor [security] (done) | time=8ms
01:58:51.518 INFO  ------------ Run sensors on project
01:58:52.059 INFO  Sensor JsSecuritySensorV2 [jasmin]
01:58:52.064 INFO  12 file(s) will be analysed by SonarJasmin.
01:58:58.446 INFO  Analysis progress:   8% (1/12 files)
01:58:58.453 INFO  Analysis progress:  16% (2/12 files)
01:58:58.459 INFO  Analysis progress:  25% (3/12 files)
01:58:59.060 INFO  Analysis progress:  33% (4/12 files)
01:58:59.108 INFO  Analysis progress:  41% (5/12 files)
01:58:59.140 INFO  Analysis progress:  50% (6/12 files)
01:58:59.141 INFO  Analysis progress:  58% (7/12 files)
01:58:59.150 INFO  Analysis progress:  66% (8/12 files)
01:58:59.164 INFO  Analysis progress:  75% (9/12 files)
01:58:59.603 INFO  Analysis progress:  83% (10/12 files)
01:58:59.734 INFO  Analysis progress:  91% (11/12 files)
01:59:05.148 INFO  Sensor JsSecuritySensorV2 [jasmin] (done) | time=13089ms
01:59:05.149 INFO  Sensor JsArchitectureSensor [architecture]
01:59:05.193 INFO  Sensor JsArchitectureSensor [architecture] (done) | time=44ms
01:59:05.194 INFO  Sensor Zero Coverage Sensor
01:59:05.239 INFO  Sensor Zero Coverage Sensor (done) | time=46ms
01:59:05.240 INFO  ------------ Gather SCA dependencies on project
01:59:05.622 INFO  Checking if SCA is enabled for organization hasibaltahsin
01:59:06.166 INFO  Dependency analysis skipped
01:59:06.171 INFO  SCM Publisher SCM provider for this project is: git
01:59:06.176 INFO  SCM Publisher 52 source files to be analyzed
01:59:10.091 INFO  SCM Publisher 51/52 source files have been analyzed (done) | time=3914ms
01:59:10.099 WARN  Missing blame information for the following files:
01:59:10.100 WARN    * snyk-results-dvna.json
01:59:10.101 WARN  This may lead to missing/broken features in SonarCloud
01:59:10.109 INFO  CPD Executor 3 files had no CPD blocks
01:59:10.110 INFO  CPD Executor Calculating CPD for 9 files
```

```
01:58:59.150 INFO  Analysis progress:  66% (8/12 files)
01:58:59.164 INFO  Analysis progress:  75% (9/12 files)
01:58:59.603 INFO  Analysis progress:  83% (10/12 files)
01:58:59.734 INFO  Analysis progress:  91% (11/12 files)
01:59:05.148 INFO  Sensor JsSecuritySensorV2 [jasmin] (done) | time=13089ms
01:59:05.149 INFO  Sensor JsArchitectureSensor [architecture]
01:59:05.193 INFO  Sensor JsArchitectureSensor [architecture] (done) | time=44ms
01:59:05.194 INFO  Sensor Zero Coverage Sensor
01:59:05.239 INFO  Sensor Zero Coverage Sensor (done) | time=46ms
01:59:05.240 INFO  ------------- Gather SCA dependencies on project
01:59:05.622 INFO  Checking if SCA is enabled for organization hasibaltahsin
01:59:06.166 INFO  Dependency analysis skipped
01:59:06.171 INFO  SCM Publisher SCM provider for this project is: git
01:59:06.176 INFO  SCM Publisher 52 source files to be analyzed
01:59:10.091 INFO  SCM Publisher 51/52 source files have been analyzed (done) | time=3914ms
01:59:10.099 WARN  Missing blame information for the following files:
01:59:10.100 WARN    * snyk-results-dvna.json
01:59:10.101 WARN  This may lead to missing/broken features in SonarCloud
01:59:10.109 INFO  CPD Executor 3 files had no CPD blocks
01:59:10.110 INFO  CPD Executor Calculating CPD for 9 files
01:59:10.162 INFO  CPD Executor CPD calculation finished (done) | time=53ms
01:59:16.380 INFO  Analysis report generated in 976ms, dir size=1 MB
01:59:16.842 INFO  Analysis report compressed in 419ms, zip size=308 KB
01:59:17.939 INFO  Analysis report uploaded in 1097ms
01:59:17.944 INFO  ANALYSIS SUCCESSFUL, you can find the results at: https://sonarcloud.io/dashboard?id=hasibaltahsin_dvna
01:59:17.945 INFO  Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
01:59:17.946 INFO  More about the report processing at https://sonarcloud.io/api/ce/task?id=AZnkTkjGQA1F3N1ntDk6
01:59:17.949 INFO  ------------- Upload SCA dependency files
01:59:20.038 INFO  Sensor cache published successfully
01:59:20.646 INFO  Analysis total time: 1:55.511 s
01:59:20.650 INFO  SonarScanner Engine completed successfully
01:59:21.202 INFO  EXECUTION SUCCESS
01:59:21.204 INFO  Total time: 2:06.783s
```

Finally, the success of Execution was seen in the output.

Viewing SonarCloud Reports (GUI)

First, I opened SonarCloud in the browser. Then, I went to my Organization and clicked on Projects - HASIB AL TAHSIN-dvna. Then, in the User Interface (UI) of SonarCloud, various analysis indicators are displayed, such as - Security, Reliability, Maintainability, Duplications, and Coverage, etc.



References

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[2] M. Ouwehand, "Please provide compiled classes of your project with sonar.java.binaries," *Stack Overflow*, Mar. 25, 2020. Available: https://stackoverflow.com/questions/46976567/please-provide-compiled-classes-of-your-project-with-sonar-java-binaries.

[3] G. Ann Campbell, "[sonar-scanner error] sonar.java.binaries property," *Sonar Community*, Apr. 9, 2019. Available: https://community.sonarsource.com/t/sonar-scanner-error-sonar-java-binaries-property/8602.

[4] "Java | SonarQube Docs," *SonarQube Documentation*. Available: https://scm.thm.de/sonar/documentation/analysis/languages/java/.

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[7] "SonarQube step configuration," *Harness STO Documentation*. Available: https://developer.harness.io/docs/security-testing-orchestration/sto-techref-category/sonarqube-sonar-scanner-reference/.